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Review article

Enhancing photovoltaic systems: A comprehensive review of cooling, concentration, spectral splitting, and tracking techniques

Mahmoud M. Abd-Elhady*, Mohab A. Elhendawy, Muhannad S. Abd-Elmajeed, Rahaf B. Rizk

Mechanical Power Engineering Department, Faculty of Engineering, Damietta University, New Damietta, Damietta, Egypt

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ABSTRACT

As the demand for renewable energy sources continues to rise, the integration of solar photovoltaic technology has emerged as a promising solution. But since these panels benefit from only about 15–20% of the radiation falling on them, the current trend has been towards improving the performance of these panels and raising their efficiency to benefit from the greatest possible amount of solar radiation. In this regard, this review presents a thorough examination of the latest advancements in enhancing photovoltaic (PV) systems, with a focus on cooling techniques, concentration methods, spectral splitting filtering, and tracking systems. A comprehensive analysis of both passive and active cooling technologies was presented, where cooling improves the efficiency of the panel to up to 25%, in addition to the possibility of integrating PV modules with thermal collectors to exploit the wasted heat in what is known as Photovoltaic/thermal systems (PV/T). Additionally, the study explores various types of concentrated PV (CPV) systems, such as flat concentrators, compound parabolic concentrators, and Fresnel lenses, which can lead to 10% increase in the panel efficiency. The paper also introduces the concept of spectral splitting filtering, covering reflective and absorptive methods to optimize solar energy utilization, which is considered a new technology cooling method. Furthermore, it delves into tracking systems for PV modules which can cause an increase in efficiency about 20%. The novelty of this review lies in its up-to-date coverage of the latest research and advancements in PV system enhancement. There are several studies that deal with a specific modification to improve the performance of the PV panel, while this study provides a comprehensive overview of the state-of-the-art techniques including four different types of modern modifications and analyzes them extensively to reach the best possible configuration, as this review aims to contribute to the continued development of effective and sustainable photovoltaic systems.

1. Introduction

Energy has been a treasure, and the most crucial prerequisite for human existence from the very first days of civilization treasures; its importance has not diminished in the modern era. With the recent technological advancements and better living conditions for the population, the demand for this factor of existence has significantly grown [1]. Thus, the search for alternative ways of ensuring the right amount of energy, particularly renewable sources, has received a major push. In the past generations, the sphere of exploration mostly depended on non-renewable sources, as it was especially noticeable in the very days of the first industrial revolution when thermal and electrical energy was produced by engines running on fossil fuels. However, as such energy tends to have too high an impact on the natural environment and is threatened to run out, humanity recently began to seek an alternative way of producing energy. The ecological disturbances that non-renewable energy has contributed to continue to urge the transition to more usable sources [2]. Considering the current issue of climate change and the growing levels of environmental contaminants, the world has developed a focus on generating energy with environmentally friendly carbon emissions [3,4].

Solar photovoltaic (PV) technology has received immense global demand as an alternative source for electric power generation. Solar energy manufacturing has witnessed a remarkable expansion over the previous decade, as it has increased tenfold globally to meet the growing demand for clean energy. It is also expected that the rate of increase of solar energy manufacturing will be about 40% until 2030 [5]. PV panels convert solar radiation into an electric current without the use of moving parts [6]. It utilizes solar cells as conversion devices or panels that convert sunlight directly into an electric current [7]. PV technology itself was first launched in 1883 when Charles Fritts produced the first panel with little performance of about 1% [8]. Since then, extensive research spanning several decades has been conducted to enhance the efficiency of photovoltaic systems. PV panels

* Corresponding author.
E-mail address: m.abdelhady@du.edu.eg (M.M. Abd-Elhady).

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Nomenclature

A_a	Aperture area [m^2]
A_r	Receiver area [m^2]
A_R	submerged area ratio [m^2]
I_{sc}	Short circuit current [A]
P_m	Peak power [W]
V_{oc}	Open circuit voltage [volt]

Abbreviations

AC	Alternating current
CNT	Carbon nano tube
CPV	Concentrated PV
CPC	Compound parabolic concentrator
CR	Concentration ratio
DAST	Dual-axis solar tracker
DC	Direct current
DCCL	Dual closed-loop control
DOF	Degree of freedom
FPV	Floating Photovoltaic
GT	grid-tied
HCS	High concentration system
HRES	Hybrid renewable energy system
LCS	Low concentration system
LDR	Light-dependent resistor
MCS	Medium concentration system
MF	Micro-fins
MJSC	Multi-junction solar cell
MPPT	maximum-power-point tracking
ND	Nano disbanded
NF	Nanofluid
nMPPO	near-maximum power-point operation
OPV	On ground PV
PCM	Phase change material
PG	propylene glycol
PI	Proportional Integral
PSC	polymer solar cell
PV	Photovoltaic
PV/T	Photovoltaic-thermal
PV–TE	Photovoltaic-thermoelectric
PVSC	perovskite solar cell
SBS	Spectral beam splitting
SPSTC	semi-passive solar tracking concentrator
STS	solar tracking system

Chemical elements and compounds

Ag	Silver
AL_2O_3	Alumina
Au	Gold
CdTe	cadmium telluride
CuO	copper oxide
Cu_9S_5	Digenite
GaAs	Gallium arsenide
SiC	carbide silicon
SiO_2	Silicon dioxide
Ta_2O_5	Tantalum pentoxide
TiO_2	Titanium dioxide
ZnO	Zinc oxide

were first published in 1953 at Bell Labs, where semiconductor materials were used to convert sunlight into electricity [9]. Currently, the typical commercial conversion efficiency of PV panels ranged from 15 to 20% [10]. Solar cells with monocrystalline and polycrystalline materials account for nearly 91% of the solar panel market [11].

Despite the remarkable spread of photovoltaic panels, they face some operational problems, as they require large areas for installation to provide the necessary energy, in addition to not capturing a large amount of sunlight due to their fixed position, as well as the surrounding conditions. High temperatures also represent a major challenge that significantly reduces the efficiency of photovoltaic cells. It has been maintained that a 1 °C increase in the operational PV panel temperature reduces its electrical efficiency by 0.4–0.5% [12], which shortens the life of the panels. In this context, photovoltaic thermal (PV/T) technology emerges as one of the prospective proposed solutions [13], which is an integration between a PV module and a thermal collector. This is particularly significant, especially in arid regions, where optimal exploitation of the system is crucial.

The review process for this review followed an organized approach to ensure its comprehensiveness and credibility. First, a comprehensive search was conducted across reputable and reliable databases, to identify the targeted literature from the recently published articles to ensure their accuracy and relevance. then, the initial list of articles was screened. To select the most important and latest articles relevant to the topic under review and exclude articles failed to meet methodological standards. Finally, the selected articles underwent a detailed analysis and synthesis to extract the methodologies and key findings.

Previous important reviews have been published in this field covers one of the ways to improve the system such as [14,15], and [16] in different methods of cooling the panel, [17] in concentration types, and [18] in tracking systems and their effect on the system.

This study focuses on four main pathways that are among the most important ways to enhance system performance (Fig. 1), contrary to the usual in terms of studying a specific improvement method, which makes it more comprehensive and facilitates access to the largest possible amount of information related to this field. The first section

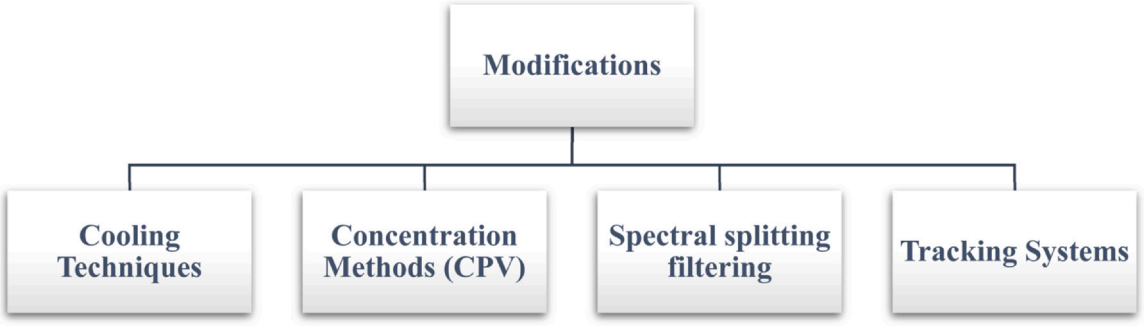


Fig. 1. Main modification pathways.

illustrates the main cooling techniques including liquid, gas, and phase-change material-based cooling. In the second section, different geometries of concentrated photovoltaics (CPVs) are analysed and characterized. The third section shows the spectral splitting techniques and how can it be used to improve the panel efficiency. Then the fourth section discusses the various categories of tracking systems. Finally, the conclusion includes a summary of what was discussed and reached to increase energy production and raise the efficiency of the PV system. The purpose of this review is to know the best configuration of the system that improves efficiency and maximizes the benefit of Solar radiation, provide recommendations, study ways in which the performance of these systems can be enhanced to be more efficient, and provide guidance for future related research work.

2. Cooling techniques

PV panels utilize sunlight to generate electrical power still, heat presents an essential challenge as it decreases panel efficiency and shortens its lifetime. It is stated that for every 10 ˚C rise in the panel temperature, its rate of degradation doubles [15]. To address this challenge, the proposed solution implies integration between the PV module with a thermal sink, inventing an integrated PV/T system. This system incorporates a heat exchanger as a thermal collector with various cooling media and their different effects [19], and the various configurations of this collector will be discussed in detail within the following section based on the cooling method. These cooling methods have been categorized into two types as illustrated in Fig. 2.

2.1. Passive cooling techniques

Passive cooling is an innovative method to cool down the PV panels without consuming energy which makes it a cost-effective, sustainable, and simple technique for cooling.

2.1.1. Passive gas cooling

This technique depends on natural convection phenomena, air (gas) is uses as the cooling medium without any driving power. In a study by Popovici et al. [20], used a heatsink was used in the back face of a PV Panel to decrease the panel temperature. Different rib heights of the heat sink (0.01:0.05 m) and different Rib Angles (45-90-135) are tested theoretically to find the best geometry for maximum efficiency. It was found that for the studied case the PV panel got a temperature of 56 ˚C if no ribs were used, and the maximum power was 86% of the nominal value. While using the heatsink, the temperature decreased by more than 10 ˚C and obtained above 90% of the nominal power. El Mays et al. [21] experimentally studied the effect of using an aluminium finned plate in the back of a PV panel with natural convection cooling. The conventional PV panel was working at a temperature that reached 56 ˚C, while after adding the finned aluminum plate, the temperature decreased by 6.1 ˚C resulting in an augment in electrical efficiency by 1.75%.

Hernandez-Perez et al. [22] studied experimentally and theoretically a new proposed configuration of a finned plate that gives a higher efficiency than the conventional finned plate. This heatsink consists of discontinuous segmented fins as shown in Fig. 3. The study tested several airflow inlet angles, and the new fin design gave the heatsink a better hydraulic performance with a reduction in the air pressure loss. The result shows that using this fin configuration makes the temperature decrease by 5–7 ˚C and the electrical efficiency increases by an average of 2.96%.

Kazem et al. [23] reviewed and examined different cooling methodologies (Passive cooling and active cooling) including air, water, heat sinks, fins and PCM, However, the authors focused on various fins configurations (passive cooling) in terms of types, design and modelling. It's found that using fins was able to increase the PVs efficiencies (electrical and thermal) by 8–10%, depending on the fin configurations.

2.1.2. Passive liquid cooling

In this technique, different types of liquid are used as heat transfer fluids, and without using any energy. For Example, Chandrasekar and Senthilkumar [24] developed a cooling system consisting of fabricated aluminium heat spreaders, cotton wicks, and headers to investigate the effect of cooling on a PV polycrystalline panel experimentally. The experiment setup was operated with a flat 25 Wp PV panel in Tiruchirappalli, Tamil Nadu, India. It was found that the maximum temperature of the unit was decreased from 49.2℃ to 43.3℃ which is about 12%, and the electrical yield was increased by 14%. Elminshawy et al. [25] made a finned U-shaped heat sink attached to a partially submerged floating PV module, with a 20% optimal submerged area ratio as shown in Fig. 4. The results show that this configuration decreased the average panel temperature by 14.93% from the conventional PV and increased the efficiency by 22.24%.

Nisar et al. [26] presented prototypes of floating PV and on-ground PV systems, each system containing two PV panels one monocrystalline and the second polycrystalline as shown in Fig. 5. The system is tested at different tilt angles (0˚, 15˚, 30˚). The average temperature of both mono and polycrystalline floating PV modules was lowered by 2–4% on the front, and 5–11% on the back compared to the on-ground PV. Thermal imaging revealed that at 0˚ tilt the cooling effect benefits more than at 30˚ tilt. Compared to on-ground PV, about 27–35% more power is produced by floating PV at 0˚ tilt, and only 8% more power is produced by floating PV at 30˚ tilt.

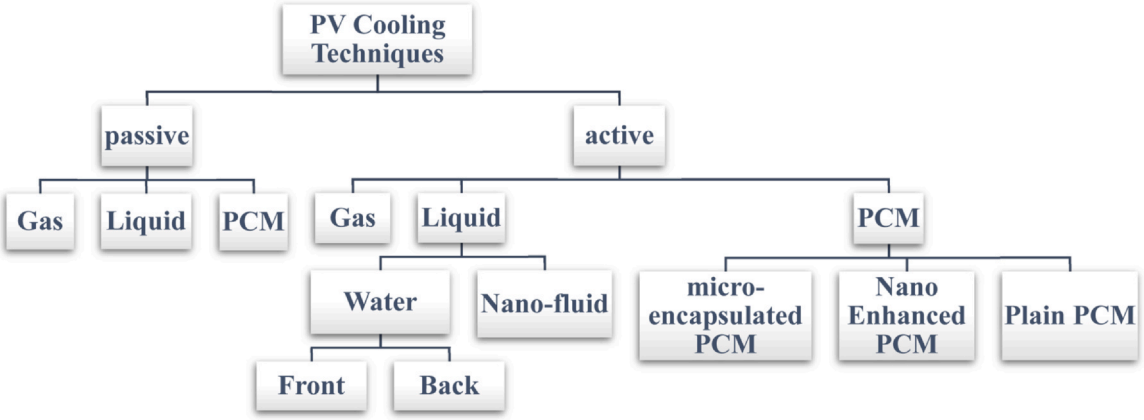


Fig. 2. Cooling methods classifications.

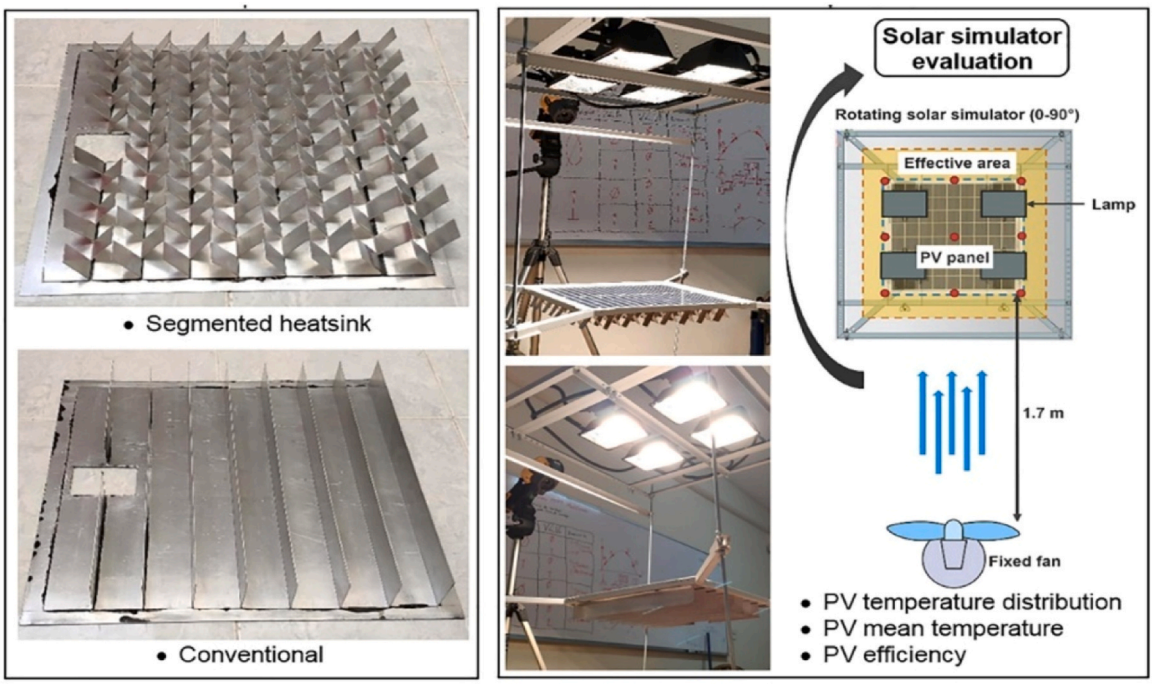


Fig. 3. Experimental methodology of Hernandez-Perez et al. study. [22].

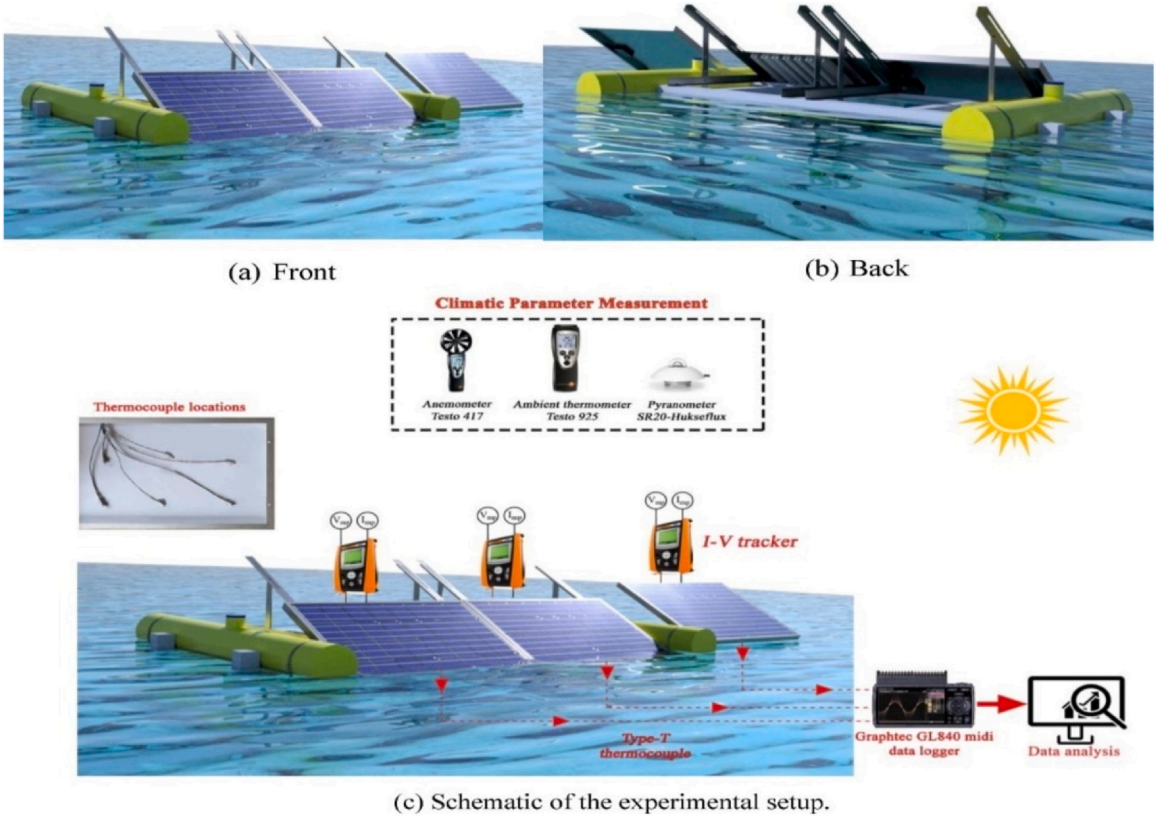


Fig. 4. Experimental setup representation of floating PV module [25]. PV = photovoltaic.



Fig. 5. Experimental configuration of floating and on-ground PV panels [26]. PV = photovoltaic.

2.1.3. Passive cooling with phase change material

Phase change materials cooling technique is a high potential method which can be used passively to absorb the waste heat of the PV system in addition to store this thermal energy. Applying phase change material (PCMs) are usually applied as thermal insulator, as no additional energy is required for operation. In a study conducted by Al Miaari and Ali [27], it was shown that using Phase Change Material (PCM) in the system for passive cooling of PV panels increased their power performance under Saudi Arabian Conditions by 5.23%. Utilizing a PCM with the correct melting point will increase the effect of cooling the PV panel while using an inappropriate PCM will reduce the panel's power output. The authors used six small containers filled with PCM instead of one single container to make it more technically applicable for practical.

Özbaş [39] demonstrated the use of thermosyphons in a passive cooling technique to increase the efficiency of PV panel by comparing two designs: the first is to use paraffin wax alone and the second is to submerge thermosyphons in paraffin. According to experimental results, the average surface temperature of the front glass was reduced to 53.6 °C, 50.5 °C, and 49.5 °C in the designs using paraffin wax alone, paraffin with methanol thermosyphons, and paraffin with water thermosyphons, respectively, compared to 54.9 °C in the reference panel. The best efficiency enhancement was achieved with paraffin with water thermosyphons improving the efficiency of the reference panel by about 6%.

Duan [28] showed how the cooling efficiency of PCM-porous systems for PV panels changes with inclination angles and Rayleigh numbers. It found that systems with larger porosity ($\varepsilon = 95\%$) are less affected by inclination angle compared to those with small porosity ($\varepsilon = 85\%, 90\%$). Weak natural convection negatively impacts melting, particularly in systems with small porosity. However, this effect extends the cooling duration for PV panels, with $\varepsilon = 85\%$ or 90% systems lasting twice as long as $\varepsilon = 95\%$ systems. Feng et al. [29] showed the potential of using nanoemulsions of low supercooling phase change material as coolants to manage the temperature of PV panels. Specific heats of PCM with low-supercooling nanoemulsions are 2.29–3.29 times water at the range of phase transitions. Using low-supercooling PCM with 20 wt% led to a reduction in the temperature by 5.3% compared to using water. Additionally, when using OP35E-nanoemulsions with 10 wt% and 20 wt%, it achieved the same temperature reduction as water-cooling systems with less than 50% of the pumping power. Table 1 summarizes some of the previously investigated PV passive cooling systems.

2.2. Active cooling techniques

Active cooling techniques depend on circulating the cooling fluid in tubes or nozzles or heat transfer systems with forced convection. Consequently, such configurations require external power and additional machine systems such as pumps to circulate the cooling fluid through the system and reduce the PV panel temperature which improves electricity production and system efficiency.

2.2.1. Gas active cooling

It relies on using gases as a heat transfer media for forced convection to enhance the heat dissipation from the PV panel and decrease the temperature. For instance, Teo et al. [10] developed an experimental setup including multiple parallel ducts with a manifold for evenly spreading airflow from a blower. An optimal mass rate of 0.055 kg/s was used to examine the system performance using air coolant and to compare it with the system without cooling. The results showed that in the absence of cooling, the temperature attained 68 °C, causing a decrease in efficiency of about 8–9%. However, with the implementation of cooling, the temperature could be lowered to 38 °C, increasing efficiency by 12–14%. Bambrook and Sproul [30] used a PV/T collector directed to the north with a 34° tilt angle at the University of New South Wales in Sydney, Australia. They investigated the effect of air mass flow rate through a fan on the system performance. The system consists of frameless PV modules embedded in a rectangular insulated duct with four channels added in the duct below the PV/T module (to provide support and act as fins), and an induced fan. It was concluded that the

Table 1
Comparative study of some previous PV passive cooling system investigations

Reference	Study type	Work target	Cooling medium	Main findings
Popovici et al. (2015) [20]	Theoretical	Investigate the impact of using air-cooled heat sinks on reducing the operating temperature of PV panels.	Gas (Air)	- The PV temperature decreased by more than 10 °C. - The maximum power produced increased from 6:39–7.55%.
El Mays et al. (2017) [21]	Experimental	Enhance the performance of PV panels using Aluminium finned plate, and cooling under natural convection.	Gas (Air)	- The efficiency of electrical conversion increased by 1.75%. - The temperature was reduced by 6.1 °C.
Hernandez-Perez et al. (2020) [22]	Theoretical	Enhance the performance of PV panels using a discontinuous finned heatsink.	Gas (Air)	- The PV panel temperature decreased by 5:7 °C. - The electrical efficiency increased by 2.96%.
Chandrasekar and Senthilkumar (2015) [24]	Experimental	Design a cooling system to investigate the cooling effect on a polycrystalline PV panel.	Liquid (Water)	- The maximum temperature decreased from 49.2 to 43.3 °C - The electrical yield increased by 14%.
Elminshawy et al. (2022) [25]	Experimental	Evaluation of the efficiency of floating PV systems with fins-assisted passive cooling.	Liquid (Water)	- The PV panel temperature decreased by 14.93%. - The power output increased by 24.02%. - The efficiency increased by 22.24%.
Nisar et al. (2022) [26]	Experimental	Comparing the On Ground PV (OPV) system with a Floating PV (FPV) system. The system content mono and polycrystalline PV panels.	Liquid (Water)	- The average temperature of both FPV panels was lowered by 2–4% on the front, and 5–11% on the back compared to the OPV. - 27–35% more power is produced by FPV at 0° tilt, and only 8% more power is produced by FPV at 30° tilt.
Al Miaari and Ali (2023) [27]	Experimental	Investigate the effect of phase change material on PV panel temperature.	PCM (RT-42)	- Using PCM reduce the PV average temperature by 10 °C. - Enhances the power performance by 5.23%.
Duan (2021) [28]	Theoretical and Experimental	Investigate the cooling effects of PCM combined with metal foam on inclined PV panels.	PCM (coconut oil)	- The cooling duration for PV panels, with $\varepsilon = 85\%$ or 90% systems lasting twice as long as $\varepsilon = 95\%$ systems.
Feng et al. (2021) [29]	Experimental	Study the thermal management performance of low-supercooling PCM nanoemulsions as coolants for PV panel systems.	PCM (OP35E nano-emulsions)	- 15–18.4% improvement in the maximum power output of the PV when using 10 wt% and 20 wt% - PCM with 20 wt% led to a reduction in the temperature by 5.3% compared to using water.

PCM = phase change material; PV = photovoltaic.

efficiencies increase as the airflow rate increases. Also, it was found that the thermal efficiency ranges between 28% and 55% and the electrical efficiency ranges between 10.6% and 12.2%.

Hussien et al. [31] studied experimentally and numerically the cooling effect on PV panels. Three different configurations have been designed for PV modules: An uncooled panel as a reference panel, an Air-cooling panel with a blower, and an Air-cooling panel with small symmetrical fans. The results showed that a system equipped with distributed fans is better than a system with a blower. The temperature decrease was about 16.7% for the fans system and 9.1% for the system with a blower and improved the output power by 22.4% and 13.2% respectively. Ooshaksaraei et al. [32] studied experimentally and theoretically four different configurations of air-based bifacial PV/T collector (single-path, double-path parallel flow, double-path counter flow and single-path returning flow). The study showed that the double-path parallel flow model has the highest energy efficiency 28–49%, and it is recommended if thermal energy was the primary output desired. Also, it was concluded that the single-path model has the highest exergy efficiency 8.2–8.4%, and it is recommended if electricity was the primary output desired. Li et al. [33] designed a compressed air system for an array of PV panels and studied the effect of this compressed air on cooling and cleaning the panels simultaneously which finally improved efficiency. It was found that by changing the time step of air blowing between three different values of 10, 15, and 20 seconds, the output power of the system increased by 30.7%–33.6% and 36.1% respectively for the three-time steps compared with the conventual case which produced 567.4 W. The authors also mention that although releasing airflow can cool the panel's surface and increase power after cleaning, the return on investment will decrease. The effect of cooling can only be maintained for a shorter time than the effect of cleaning, so the energy used to pump the required flow rate of air was higher than the energy gained from increasing the time of cooling.

2.2.2. *Liquid active cooling*

Liquid-based cooling involves liquid coolants such as water and other specialized fluids used to cool the temperature of PV panels, carrying away the heat and enhancing the electrical and thermal efficiency. Forced circulation is applied to increase the heat dissipated from the panel. When a liquid-based cooling system is compared to air cooling, hydraulic cooling in a PV system provides more electrical power, thermal efficiency, and total efficiency.

2.2.2.1. Water active cooling. Water is the most commonly used liquid as a coolant in PV/T systems. Water cooling is added to the photovoltaic system to reduce the temperature of the panels to enhance electrical production capacity, in addition to obtaining thermal energy which was wasted and harmful to the panel by reusing the hot water resulting from cooling in other applications [34]. Several technologies of active water cooling have been presented; the first approach is surface cooling by spraying water on the front surface of the panel. In addition to its primary function, which is lowering the panel's temperature, this technique helps in cleaning the panel and reducing dust deposition on it.

It is also possible to combine cooling with water and air, as a study by Al-Waeli et al. [35] to review and analyse different configurations of bi-fluid PV/T collectors that use different modes such as (air only, water only, dual fluid).

Kazem et al. [36] conducted an outdoors experimental in Energy Lab at Sohar University, Oman. Where the PV/T hybrid collector (oscillatory/direct) with three PV/Ts (Web, direct and oscillatory) and conventional PV were designed and tested, All PV panel for the PV/Ts have the same specifications of the conventional PV, which is 100 W, 5.94 A, and 22.32 V, and a 0.020 kg/s mass flow rate was used. the results showed that hybrid PV/T outperforms the other three PV/Ts, it has 9.11%, 9.98%, and 15.28% higher efficiency than oscillatory, direct and web, respectively. And it achieved higher voltage and power ± 18 V and 64.6 W compared with ± 14 V and 54.1 W of the conventional PV.

Krauter [37] was the first patent to investigate the effect of sprayed water on the front side of a PV panel to enhance its electrical efficiency. A water film of 1 mm thickness was applied at the front side at a flow rate of 2 L/min through 12 nozzles placed at the top of the panel. It was found that this method decreased the temperature of the PV panel by 22 °C when compared with conventional PV and increased the electrical energy by about 10.3%. Moharram et al. [38] conducted an experimental study on a closed-cycle water cooling system designed for PV panels. The cooling system integrates a suction pump that draws water from a supply tank. The water goes through a filtration process before being converted into a thin film and sprayed over the panel. Subsequently, the water is collected and returned to the tank, completing the closed cycle. By using the closed cycle, this configuration significantly reduces water consumption, which is of utmost importance in arid regions. It was found that when the cooling system operates for five minutes, a temperature reduction of 10 °C can be achieved and an increase in the conversion efficiency of the solar cell by 12.5%. Smith et al. [39] operated an experimental investigation to enhance the electrical output of a low concentration photovoltaic (LCPV) system by implementing front-side cooling. A water film was sprayed over the front side of the PV panel, then collected, and recirculated. This cooling cycle effectively reduced the panel temperature from 55 to 40 °C, resulting in a power enhancement of 9.4%.

In the study by Matias et al. [40], the performance of PV panels was examined to regulate the optimal volume flow rate for module cooling. A cooling system with a luminaire was installed to simulate solar radiation. Several water flow rates ranging from 0.5 to 4 L/min were tested. The experiments revealed that the reduction in panel temperature from 70 to 50 °C, was accompanied by an increase in power output from 62 to 77 Wh. Furthermore, it was found that the initially estimated flow rate was overvalued, and the optimum flow rate was determined to be 0.6 L/min, causing a net energy gain of 77.41 Wh. This represents an enhancement of about 22.7% compared to the panel without cooling. Hadipour et al. [41] designed a pulsed-spray cooling system employing water to enhance PV performance while decreasing water consumption. The results were compared with steady spray and non-cooled systems. The study revealed that the steady spray cooling system achieved the greatest electrical power, increasing by approximately 33.3%. Subsequently, the pulsed-spray cooling system with a duty cycle (ratio of on-time/off-time/cycle) of 1 and 0.2 was tested, demonstrating an insignificant decrease in efficiency to 27.7% and 25.9%, respectively.

Another approach to cooling PV panels is through the back-side cooling, where water flows either through tubes or is sprayed on the back of the panel. The most popular design for PV/T collectors presents an absorber made of metallic sheets and tubes attached to the back of PV cells, with a forced flow of water. Copper is a common material utilized due to its excellent heat conductivity and other beneficial properties [42]. Kim and Kim [43] investigated the electrical and thermal efficiencies of a water-type PV/T collector featuring a fully wetted absorber. They built a glassless PV/T collector prototype, tested it outdoors in a location in South Korea, and compared it to a conventional PV panel. The results demonstrated that the overall efficiency of the wetted absorber system improved to 65%, resulting in enhanced electrical and thermal output. Also, it was found that when the glassless PV panel was installed outdoors and exposed to the surrounding air, its electricity efficiency was about 12%. Additionally, while using the fully wetted PVT collector, the efficiency increases by approximately 2% compared to the PV panel alone. Generally, the PVT's electrical efficiency has improved by 15% due to the cooling.

Alzaabi et al. [44] performed an experimental study on a polycrystalline PV panel with a thermal collector to improve PV panel performance. A parallel tube flat plate collector made of rectangular copper pipes is attached to the back of a 100-W PV panel. The rectangular shape of the pipes helps increase the contact area between the

collector and the PV panel. The authors developed a PV/T system prototype and evaluated its electrical and thermal efficiencies under ambient temperature conditions in Sharjah, United Arab Emirates in April 2014. The results revealed a 5% increase in electrical power output for the PV/T system, achieving 20% compared to a conventional PV panel. Furthermore, thermal efficiency obtained an advancement of 60–70%.

Aste et al. [45] conducted a comprehensive numerical and experimental investigation of back-cooling PV/T systems. The tested PV/T system comprised a PV module containing 30 polycrystalline cells in a series connection. These cells were mechanically linked to a roll-bond aluminum heat sink, as illustrated in Fig. 6a. The PV/T system was oriented towards the south with a tilt angle of 30°. The thermal cycle consists of a 200-litre insulated storage tank, pipelines, and a pump, as shown in Fig. 6b. The experimental work spanned two years, from 2013 to 2015, and was conducted at Politecnico di Milano. Outdoor experiments were carried out to validate and compare with the simulation data. The experiments were performed in clear-sky weather conditions and on rainy days, with a cooling water flow rate of 0.055 kg/s. The overall efficiency increased based on the annual radiation levels, reaching 32.7% in Paris, 36.1% in Milan, and 40.6% in Athens. However, the electrical efficiency decreased with increasing average module temperature, resulting in 13.7% in Paris, 13.6% in Milan, and 13.4% in Athens.

Rahou et al. [46] investigated the performance of a PV/T system using an oscillatory flow design. The system consisted of a rear plate insulated with glass wool and an integrated structure of oscillatory flow copper tube heat sinks. A single manual tracker was employed. The total efficiency, comprehending both electrical and thermal efficiencies, ranged from 70.53 to 81.5% for mass flow rates varying from 1 to 3.5 L/min, with an inlet temperature of 30 °C. Nahar et al. [47] conducted experimental and simulation study on the performance of a PV/T system with a parallel plate flow channel. A finite element approach was devoted to a 3D numerical analysis of the PV/T system using COMSOL software. The results showed that the numerical and experimental highest total efficiency of the PV/T system was 84.4% and 80% for solar radiation level of 1000 W/m^2 and intake and ambient temperatures of 34 °C.

2.2.2.2. Nano-based active cooling. The utilization of nanoparticles in a liquid to increase the conductivity of the working fluid is a cooling technique used to enhance the overall efficiency of a system. Abdallah et al. [48] conducted a study applying Al_2O_3 nanofluid with variable concentrations (0.05%, 0.075%, 0.1%, 0.2%, and 0.3% V) as a coolant to a PV system. A comparison was made between water-based PV/T and Al_2O_3 nanofluid-based PV/T. The optimal concentration was determined to be 0.1%, causing an average temperature reduction of 8.6 degrees throughout the day with a 24.6% increase in electrical efficiency. The total efficiency attained about 56.16% over the course of the day. Under maximum solar radiation, the temperature decreased by about 10°, complemented by a 26.3% increase in electrical efficiency and a total efficiency of 74.17%. Al-waeli et al. [49] examined the performance of PV/T system by using PCM and nano cooling system and compared them with mathematical models and artificial neural networks to reduce the temperature of PV cell. The results were satisfactory for the best model as the coefficient of determination (R^2) score was 0.81, mean square error was 0.0361 and root mean square error was 0.371.

In a study by Tyagi and Kumar [50], they developed and fabricated an outdoors experimental set-up (PV/T) system (thermal collector with a multi-block finned absorber plate) in India. the authors proposed a novel system that combines active cooling (multi-walled carbon nanotubes (MWCNT) - water nanofluid) and passive cooling (paraffin wax as the PCM) techniques for thermal regulation in so-called PV/T-PCM. It was found that the PV/T-PCM average and maximum reductions in panel temperature ranged from 10.9 to 14 °C and 15.9 to 20.1 °C compared to a conventional PV module. results also showed that average percentage increase in energetic electrical efficiency was 7.57–9.19%.

Praveen and Rajan [51] established a thermodynamic model of a nanofluid based PV/T system with novel double-loop rectangular spiral collector design, to validate a programming code that demonstrates an iterative resolution analytical method. The following four operating parameters were used to predict the performance, solar irradiation (G = 300–900 W/m^2), ambient temperature (Ta = 283.15–313.15 K), CuO nanoparticles concentration (φ = 0.02–0.08%) and mass flow rate of nanofluid ($\dot{m}_{nf}$ = 201–241 kg/h). The recommended values were G = 855.353 W/m^2, Ta = 313.148 K, φ = 0.08%, and $\dot{m}_{nf}$ = 201 kg/h, as these values predicts the highest output with 78.981 W energetic electrical power, 320.075 W energetic thermal power, 78.981 W exergetic electrical power, and 0.953 W exergetic thermal power.

Menon et al. [52] a comparison was presented between water-based PV/T and nanofluid-based PV/T utilizing CuO nanofluid. The mass flow rate of both cooling fluids was maintained at 0.067 kg/s. Water-based PV/T demonstrated a temperature reduction of 15 °C, along with 12.32% higher electrical efficiency and 58.77% higher thermal efficiency. On the other hand, nanofluid-based PV/T achieved a temperature reduction of 23.7 °C, a 35.67% increase in electrical efficiency, and 71.71% thermal efficiency. Al-Waeli et al. [53] conducted tests to measure the stability and conductivity of a nanofluid, specifically using

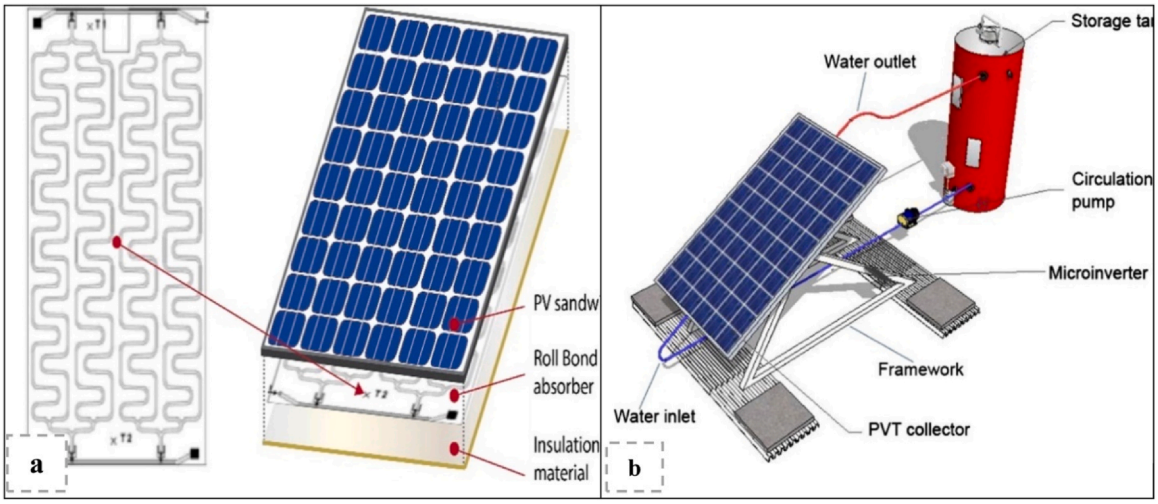


Fig. 6. (a) The PVT collector configuration. (b) The whole system configuration [45]. PVT = photovoltaic thermal.

3 wt% SiC nanofluid, in the cooling system of a PV panel. The nanofluid showed a higher electrical efficiency of 24.1% and a 100.19% thermal efficiency compared to water-based PV/T. The addition of nanoparticles led to an 8.2% increase in density and a viscosity increase of up to 5.18%. The nanofluid thermal conductivity increased by 4.3%, highlighting the importance of contemplating these factors in pump selection.

Bhat and Qayoum [54] investigated a novel pipe configuration for cooling PV panels using nanofluids (a combination of three nanofluids). The authors compared the performance of a straight tube, a wavy tube, and a hybrid configuration, aiming to detect the optimum setup. Various flow rates were also tested. It was found that the use of the hybrid tube configuration resulted in a 7.3% enhancement in electrical efficiency, a significant improvement of 16.8% in thermal efficiency, and a remarkable increase of 65% in thermohydraulic efficiency. Additionally, pressure loss was reduced by 30% compared to the wavy tube configuration. In a study by Al-Shamani et al. [55], a comparison was made between three nanofluids (TiO_2, SiC, SiO_2) and a water-cooled PV/T system, considering four different flow rates to determine the optimal configuration in terms of heat transfer fluid and flow rate. It was found that SiC nanofluid produced the highest efficiency. Conventional PV module efficiency ranged from 6.06 to 10.01%, whereas the implementation of a PV/T system with SiC achieved an electrical efficiency of 13.52% and a thermal efficiency of 68.21% with a mass flow rate of 0.170 kg/s.

2.2.3. Phase change materials-based active cooling

The incorporation of PCMs into the design of PV panels offers a means to enhance system efficiency while enabling the storage and release of thermal energy during phase transitions. PCMs are often used in complex system configurations. In a study conducted by Hamada et al. [56], an innovative PV/T system was proposed which utilized PCM balls combined with water for both passive and active cooling. The actively cooled PV/T-PCM panel demonstrated an overall efficiency of 74.1%, surpassing the values of 34.6% for the passively cooled PV/T-PCM panel and 12% for the reference PV panel. Fu et al. [57] presented a cooling system utilizing microencapsulated phase change material slurry, which resulted in a 13.5% increase in thermal efficiency and a 0.8% in electrical efficiency improvement compared to water-cooled PV/T systems. Al-waeli et al. [58] aimed to control the heat capacitance of the system and enhance overall efficiency by utilizing PCM with a cooling system that circulates nanofluids. Nano-SiC particles were incorporated into the paraffin-PCM to increase the thermal conductivity, also increasing the thermal efficiency to 72%. and improve the electrical efficiency by about 6.6%. Bassam et al. [59] applied a cooling nanofluid circulation system with a micro-fin tube encapsulated with nano PCM. The cooling fluid was water and SiC with a concentration of 0.6 vol%. It was found that the utilization of micro-fin-nano-fluid-Nano-PCM resulted in an electrical efficiency of 9.6% and a thermal efficiency of 77.5%. The new system achieved an output of 14.5 W, surpassing the output of the PV panels, which was 10.49 W. Table 2 provides a simplified comparison of some studies that presented the modifications to the PV active cooling systems.

Kouravand et al. [60] designed a combined concentrated PV/T system and examined the performance of five different scenarios. The system incorporated two aluminium plates as concentrators, with copper tubes and copper fins for heat transfer fluid circulation (water/Al_2O_3 nanofluid). The five scenarios were tested and compared to a concentrated PV/T system. The results showed that using nanofluid in conjunction with copper fins in a PCM (Paraffin Wax) container generated the highest efficiency, with an electrical efficiency of 17.02% and 61.25% thermal efficiency. the reduction in average temperature of the module decreased by about 26.6 °C.

3. Concentration methods (CPV)

In order to enhance the electrical efficiency of PV panels, one proposed approach is to increase the intensity of radiation. The primary objective of implementing a concentrated photovoltaic system (CPVS) is to replace a large number of PV panels with lower initial costs and reduce the overall panel quantity required. The analysis and characterization of different geometries of CPVs, such as LCPV, medium concentration (MCPV), and high concentration (HCPV), demonstrated in Fig. 7, are the main objectives of this section. Then there is the concentrated photovoltaic system that uses a concentration of mirrors to reflect solar rays to produce heat and uses cooling media to regulate the PV temperature at a low level. Such configurations can be very effective when added in exchange for the entire system and help lower the cost of the PV setup. All of these concentration systems help to minimize the number of panels that are needed to collect enough power by using inexpensive mirrors or lenses that amplify the intensity of light to up to 36.5% more solar radiation on PV cells; therefore, fewer panels are needed [61].

Before discussing the different configurations and types of CPV systems, it is essential to define the geometries of these concentrators and their importance for CPV systems. The concentrator geometry and concentration ratio are two main parameters which must be mentioned. Firstly, concentrator geometries are fundamental to concentrated photovoltaic thermal (CPV/T) systems, where they provide to concentrate sun rays onto small and highly efficient PV cells. Lenses and mirrors are regularly applied as concentrator geometries in CPV/T systems, with lenses focusing indirect rays onto the PV cells, while mirrors reflect and concentrate sun rays onto the described region. In addition, the concentration ratio computes the degree of sun rays' concentration achieved by geometry. In other words, the concentration ratio measures the percentage between the intensified intensity of the sun's rays and their original intensity. Thus, by adjusting the concentrator geometries to enhance concentration ratios, the energy output and efficiency of CPV systems can be maximized. The concentration ratio can be calculated by dividing the concentrator aperture area (the area where sun rays enter) by the receiver area (the area where the concentrated sun rays are captured) [62].

3.1. Low concentration

Low concentration (LC) systems can provide moderate concentration ratios between 1X to 10X focusing the light onto larger areas which results in lower irradiation intensity compared to high concentration. The main advantage of LC systems is that it does not require a tracking system or cooling methods since the concentration ratio is low. Even if there is a tracking system used, these systems are more tolerant of tracking errors and can use simplified tracking systems without the need for difficult tracking algorithms. Because of their low concentration ratio, the cost of solar cells must be taken into consideration. The LCPV systems rely on mirrors and lenses as concentrators that are cheaper than PV panels, which can achieve considerable cost savings. The low conversion efficiency, with a range of 17–20% because of the high temperature of the surface and the high material cost of PV modules, is obstructing the spread of solar energy usage worldwide to generate clean power. This issue can be handled by utilizing optical concentrators, which collect sunlight and concentrate it on a smaller area [63]. Flat Concentrators, Compound parabolic concentrators (CPC) and Fresnel Lens are the typical optical configurations of LCPV systems.

3.1.1. Flat concentrators

Flat concentrators are the non-imaging type that uses flat or V-shape surfaces to reflect and concentrate sunlight with no need for a tracking system because of their large acceptance angle which allows them to

Table 2
Comparison of some studies that reviewed the PV active cooling systems

Reference	Study type	Work target	Cooling medium	Main findings
Teo et al. (2012) [10]	Theoretical and Experimental	Design system consisting of an array of parallel ducts with a manifold to distribute the air flowing with a rate of 0.055 kg/s to investigate the difference between the results without and with cooling.	Gas (Air)	- Without cooling, the temperature was 68 °C and the efficiency decreased to 8–9%. - With cooling, the temperature reduced to 38 °C and the efficiency increased to 12–14%.
Bambrook and Sproul (2012) [30]	Experimental	Use a PV/T collector directed to the north with a 34° tilt angle and test it with different air flow rates.	Gas (Air)	- The efficiency increases as the flow rate increases. - The thermal efficiency ranges between 28% and 55% - The electrical efficiency ranges between 10.6% and 12.2%.
Hussien et al. (2023) [31]	Theoretical and Experimental	Design three arrangements PV modules: Uncooled panel as a reference, panel with blower, and panel with small symmetrical fans.	Gas (Air)	- The temperature reduction was about 16.7% for the fans system and 9.1% for the system with a blower. - Output power was enhanced by 22.4% for the fans system and 13.2% for the system with a blower.
Ooshaksaraei et al. (2017) [32]	Theoretical and Experimental	Study four configurations of bifacial PV/T collector (single-path, double-path parallel flow, double-path counter flow and single-path returning flow).	Gas (Air)	- The first model has the highest energy efficiency of 51–67%. - The second model has highest exergy efficiency of 8.2–8.4%. recommended if the thermal energy is the desired output.
Li et al. (2021) [33]	Theoretical and Experimental	design of a system of PV panels array to study the effect of cooling and cleaning the panels on its efficiency.	Gas (Compressed air)	- For the 10-15-20 seconds time step of blowing time, the output power of the system increased by 30.7–33.6 and 36.1% respectively.
Kazem et al. (2021) [36]	Experimental	Compares a hybrid collector (oscillatory/direct) PV/T with three PV/Ts (Web, direct and oscillatory) and conventional PV.	Liquid (water)	- The hybrid PV/T outperforms the other three systems, it has 9.11%, 9.98%, and 15.28% higher efficiency than oscillatory, direct and web, respectively - the hybrid PV/T higher voltage and power of ± 18 V and 64.6 W compared with ± 14 V and 54.1 W of the conventional PV
Krauter (2004) [37]	Experimental	Enhance the electrical efficiency by spraying water on the front side of the PV panel.	Liquid (water)	- The electrical energy gain was 10.3%. - The panel temperature decreased by 22 °C compared with conventional PV.
Moharram et al. (2013) [38]	Experimental	Use water cooling to improve the performance of the PV panels.	Liquid (water)	- The temperature dropped by 10 °C when the cooling system was operated for 5 min causing an increase in cell efficiency by 12.5%.
Smith et al. (2014) [39]	Experimental	Improve electric output by performing an investigation using front-side cooling for a low-concentrated PV system.	Liquid (water)	- The power enhancement was 9.4%. - The panel temperature was reduced from 55 to 40 °C.
Matias et al. (2017) [40]	Experimental	Enhancing PV panel power through water cooling technique.	Liquid (water)	- The panel temperature was reduced from 70 to 50 °C and the power increased from 62 Wh to 77 Wh.
Hadipour et al. (2021) [41]	Theoretical and Experimental	Design a pulsed-spray cooling system using water to enhance the efficiency of the PV and reduce consumed water.	Liquid (water)	- The electrical power with the steady-spray system increases by about 33.3%. - For pulsed-spray systems with duty cycles of 1 and 0.2, the efficiency reached 27.7% and 25.9%, respectively.
Kim and Kim (2012) [43]	Experimental	Compare an unglazed prototype PV/T collector with a conventional PV by testing them in outdoor conditions.	Liquid (water)	- The overall efficiency of the wetted absorber system improved to 65%.
Alzaabi et al. (2014) [44]	Experimental	Teste PV panel with a thermal collector to improve the efficiency and compare it with a conventional panel.	Liquid (water)	- The electrical power output increased from 15 to 20%. - The thermal efficiency improved by 60–70%.
Aste et al. (2016) [45]	Theoretical and Experimental	Present models of a back-cooling PV/T system with a water flow rate of 0.055 kg/s. and investigate the system under the climate conditions of Paris, Milan, and Athens.	Liquid (water)	- The overall efficiency increases to 32.7% in Paris, 36.1% in Milan and 40.6% in Athens. - The electrical efficiency decreases according to the average module temperature to 13.7% in Paris, 13.6% in Milan and 13.4% in Athens.
Rahou et al. (2013) [46]	Experimental	Study the performance of PV/T system with different mass flow rates of water from 1 to 3.5 L/min with inlet temperature of 30 °C.	Liquid (water)	- The total efficiency ranges from 70.53 to 81.5%.
Nahar et al. (2017) [47]	Theoretical and Experimental	Carry out studies on the performance of a PV/T system with a parallel plate flow channel.	Liquid (water)	- The highest numerical and experimental efficiency was 84.4% and 80%, respectively, for 1000 W/m^2 irradiation and 34 °C ambient temperatures.

(*continued on next page*)

Table 2 (*continued*)

Reference	Study type	Work target	Cooling medium	Main findings
Abdallah et al. (2018) [48]	Experimental	Compare between water-based PV/T and Al_2O_3 nanofluid-based PV/T with concentrations of (0.05%, 0.075%, 0.1%, 0.2%, and 0.3% V).	Nanofluid (Al_2O_3-Water)	- At optimal concentration of 0.1% V, max. temperature reduction was 8.6 °C throughout the day, and 10 °C under max radiation. - Combined efficiency of 56.16% throughout the day, and 74.17% under max. radiation.
Tyagi and Kumar (2024) [50]		Developed a (PV/T) system (thermal collector with a multi-block finned absorber plate) and combined active cooling with passive cooling techniques and compared the result to a conventional PV module	Nanofluid (multi-walled CNT) and PCM (paraffin)	- The PV/T-PCM average and maximum reductions in panel temperature ranged from 10.9–14 °C and 15.9–20.1 °C. - The average increase in energetic electrical efficiency was 7.57–9.19%
Menon et al. (2022) [52]	Experimental	Compare the use of water and nanofluid (CuO) on the performance of an unglazed (PV/T) system with a mass flow rate of 0.067 kg/s.	Water and Nanofluid (CuO)	- Water reduced temperature by 15 °C and increased both electrical and thermal efficiencies by 12.32% and 58.77%. - Nanofluid reduced the temperature by 23.7 °C and increased electrical and thermal efficiencies by 35.67%, and 71.71%.
Al-Waeli et al. (2017) [53]	Experimental	Investigate the effect of cooling with nanofluid (SiC) on the (PV/T) system.	Nanofluid (SiC)	- The PV/T nanofluid system had an overall efficiency of 88.9% (3 wt%), increasing the electrical efficiency to 24.1%, and thermal efficiency increased to 100.19%.
Bhat and Qayoum (2023) [54]	Theoretical	Investigate a novel pipe configuration for cooling PV panels using a combination of three nanofluids, and compare its performance with the straight tube, a wavy tube.	Nanofluids and PCM	- The hybrid tube increased the electrical efficiency by 7.3%, the thermal efficiency by 16.8%, and decreased the pressure loss by 30% compared to the wave configuration.
Al-Shamani et al. (2016) [55]	Experimental	Make a comparison between three nanofluids and water-cooled PV/T systems with four distinct flow rates.	Nanofluids (TiO_2, SiC, SiO_2)	- PV/T with SiC obtained the highest values, electrical efficiency of 13.52% and thermal efficiency of 68.21% at a mass flow rate of 0.170 kg/s.
Hamada et al. (2023) [56]	Experimental	Examine water-based PV/T systems using encapsulated polychlorinated biphenyl (PCM) balls.	PCM (Encapsulated RT35 paraffin wax ball)	- Overall efficiency of 74.1% was achieved by the actively cooled PV/T-PCM panel, and 34.6% and 12% for the passively cooled PV/T-PCM panel and reference PV panel.
Fu et al. (2021) [57]	Experimental	Investigate the impact of using the cooling system with Micro encapsulated PCM on the PV/T system.	PCM (Micro encapsulated paraffin)	- The thermal efficiency increased by 13.5%. - The average electrical efficiency increased by 0.8%.
AL-Waeli et al. (2017) [58]	Experimental	Control the heat capacitance of the system and enhance overall efficiency by utilizing PCM with a cooling system that circulates nanofluids.	PCM (Paraffin wax) with Nanoparticles (SiC)	- The electrical efficiency increased by 6.6%. - The thermal efficiency reached 72%.
Bassam et al. (2023) [59]	Experimental	Investigate the effect of nano PCM and micro-fin tube counterclockwise twisted tape nanofluid on the (PV/T) collector.	PCM (Paraffin wax) with Nanoparticles (SiC)	- The PV/T had an electrical efficiency of 9.6%, and the maximum thermal efficiency was 77.5%.
Kouravand et al. (2022) [60]	Experimental	Assessment of a finned PCM heatsink integrated with a nanofluid-based concentrating PV/T system and comparison it with CPV/T.	Nanofluid (Al_2O_3-Water)	- Nanofluid with finned PCM had the highest efficiency, 17.02%.

CNT = carbon nanotubes; CPV = concentrated photovoltaic; PCM = phase change material; PV = photovoltaic; PV/T = photovoltaic thermal.

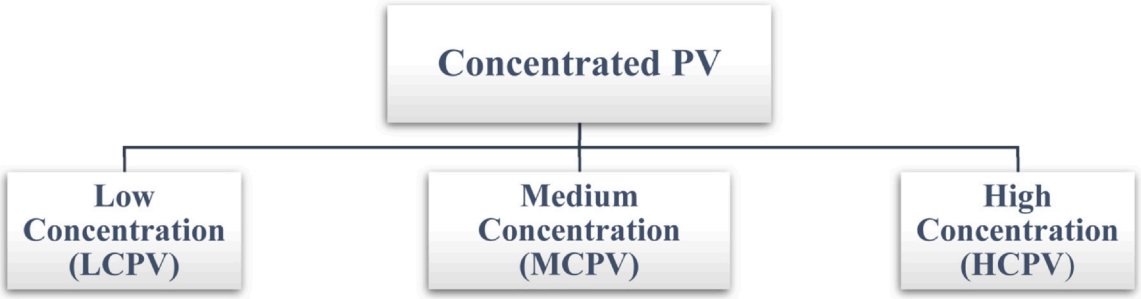


Fig. 7. Concentrated PV categories. PV = photovoltaic.

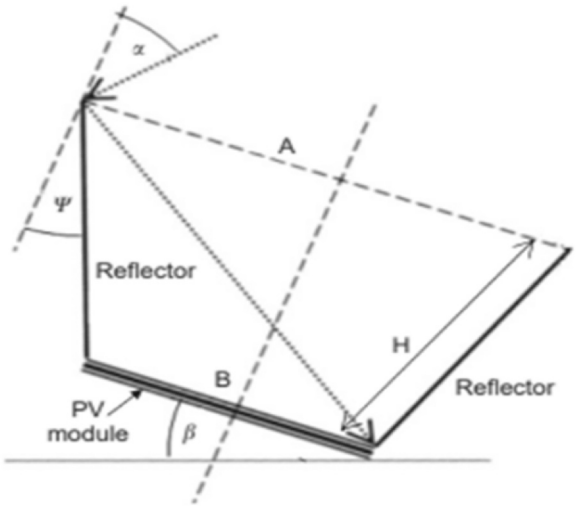


Fig. 8. Flat concentrators [64].

capture light from wider angles as shown in Fig. 8 [64]. With the advantage of low-cell heating, a flat concentrator is an excellent choice for low-cost applications reducing the complexity and cost of production.

Arshad et al. [65] proposed a method for enhancing the efficiency of solar panels that relies on experimental data acquired through three distinct approaches. An iron frame was specifically designed to accommodate the solar panel and three mirrors. While adding an active method of cooling by using water to cool down the PV temperature and enhance its performance. The experiment was repeated three times with different configurations. The first is PV without any additions, the second PV with mirrors and the third PV with mirrors and cooling. It was found that the combination of mirrors and cooling surpassed the other two configurations, with an efficiency of almost 52% while with using the mirrors only the efficiency improved by 32%. Malik and Chandel [66] tried to enhance the electrical efficiency of a PV panel by using Flat mirrors located in Hamirpur, India. The reflectors were installed at the ideal tilt angle for the designed system. The results show that the power output of the system is maximized resulting in an increment of its power potential. The power enhancement is found to be 19.84% in summer and 13.23% in winter. Also, the results show that during the summer the PV panel with flat mirrors recorded an average temperature in the range of 48–50 °C with a maximum value of 59 °C. Kolamroudi et al. [67] designed a low-concentration system with active cooling to enhance the PV module. The authors used from one to four mirrors to increase the solar irradiance with the cooling system. A reference PV panel was used along with the modified one and it was found that the DC power surpasses the reference panel's power by almost three times.

Abdullah et al. [68] investigated the impact of a reflective mirror on the performance of a Trombe wall/photovoltaic system. Reflective mirrors were employed to enhance the reflection of solar energy onto the solar panels. Additionally, fans were used to reduce PV surface temperature. The results showed that when using reflective mirrors only without fans, the highest thermal efficiency ever recorded was 74.4%. In comparison, at noon the electricity efficiency that was maximum when DC fans were used without reflective mirrors was 13.9%. In the outdoor section at Jundi-Shapur University of Technology, Dezful, Iran, Lotfi et al. [69] constructed a PV system that employs two mirrors and a PCM. The utilization of reflectors raised the temperature of the PV module, leading to a decrease in generating power, so they used PCM material (paraffin) to lower the temperature and maintain it at a modest level, so it does not affect the process. Different experiments with different tilt angles were tested to determine the optimum angle. The authors concluded that among various mirror tilt angles, the 30° angle yielded the most favorable results. Subsequently, this configuration was further examined after adding another mirror above the PV module. This double-mirror setup resulted in a 4.1% enhancement in electrical output compared to the conventional setup. Finally, the optimal configuration was determined with the addition of PCM on the backside, resulting in a significant 12.5% increase in electrical output.

3.1.2. *Compound parabolic concentrators*

In the realm of low-concentration solar PV systems, compound parabolic concentrators (CPCs) are preferred due to their superior optical efficiency, broader acceptance angles, and ability to capture both direct and diffuse sunlight [61]. Hadavinia and Singh [70] compared experimentally the performance of PV with CPC and PV with a V-trough (flat concentrator). A small-scale PV panel is used with a 50-millimeter-trimmed CPC and then with a V-trough collector with a 22° TWA (trough wall angle). The experimental findings indicated that the CPC concentrator outperformed, resulting in a 2.4% increase in power output higher than the V-trough collector.

In Alexandria, Egypt, Yousef et al. [71] performed a comparison experiment between a CPC/PV system and a conventional PV panel, in CPC/PVs as shown in Fig. 9, two 2-D CPCs were constructed. The CPC trough had dimensions of 22 cm, 59 cm, and 24.5 cm. The trough was cut in half, reducing the concentration ratio from 2.66 to 2.4. An aluminium mirror sheet was used as a reflecting surface. Experimental and numerical analyses for both configurations, with and without cooling, were demonstrated. The results show that the concentrated PV system outperformed the unconcentrated PV system for both non-cooling and cooling cases, by 33% and 52%, respectively, and Cooling reduced the temperature of the non-concentrated PV and the concentrated PV by about 25% and 30%, respectively. Consequently, the electrical power increased by 11% and 15%.

Some researchers studied the influence of scale on the electrical properties of CPC-PV systems. Li et al. [72] provided a comparative experiment between two distinct scales of CPC systems that have an identical CR of 4X. According to the findings, In the context of CPC-PV cells with two different scales but having the same geometric concentration ratio, the results showed that the small-scale CPC-PV cells

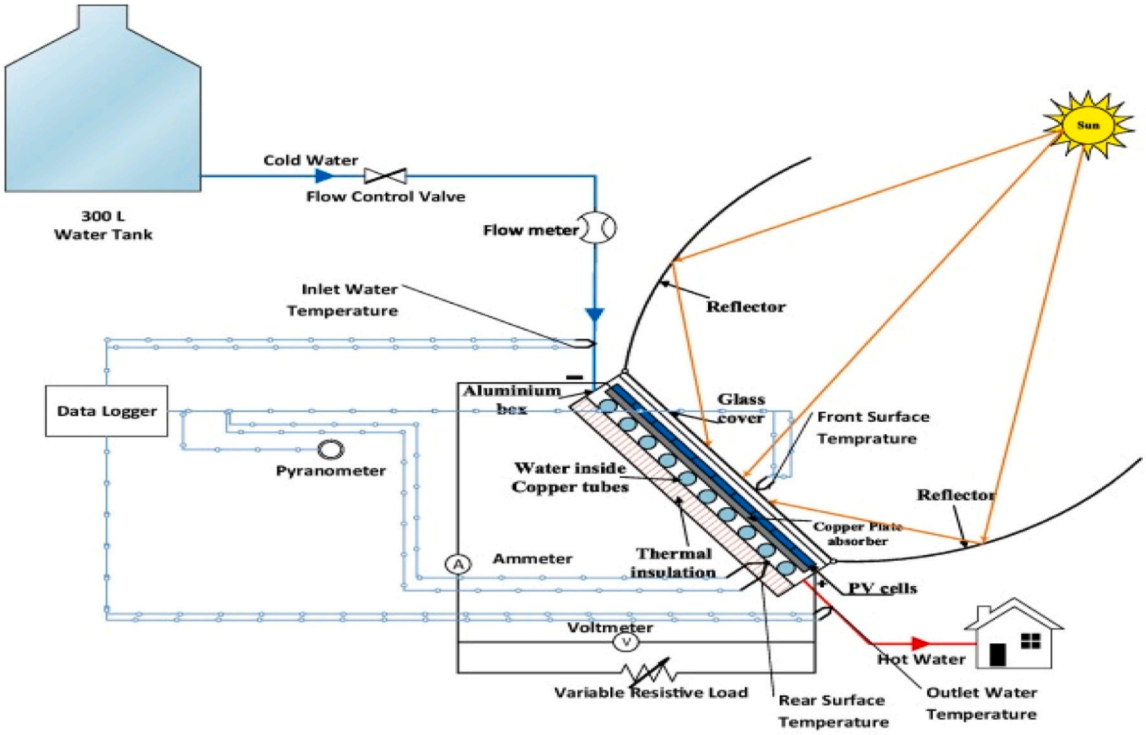


Fig. 9. Schematic diagram of a CPC-PV collector [71]. CPC = compound parabolic concentrator; PV = photovoltaic.

exhibit better electrical efficiency. It generated a higher maximum output power of 424.960 mW compared to the larger-scale CPC-PV cells, which have a slightly lower maximum output power of 420.713 mW.

Jaaz et al. [73] studied the effect of concentrating the sunlight on the PV panel using CPC mirrors. Using jet impingement to spray water on the back of the PV surface to reduce the panel's temperature as a cooling method. They built a prototype of a system installing a jet impingement and compound parabolic concentrator. At 1:30 pm, the power generation of the PV module increased by 31% while using the cooling cycle with CPC, and 16% without CPC. Therefore, even in the absence of cooling, the utilization of CPC alone can result in significant improvements in power generation and output efficiency of the PV module.

3.1.3. Fresnel lens

Fresnel lenses are optical devices that consist of a series of concentric, stepped, and flattened ridges or prisms on a flat surface. They are used as light-collecting tools as they capture the light from wider areas and focus it on smaller areas because of their great optical efficiency, in addition to their low price and lighter weight. Initially, the mainstream fresnel lenses utilized in solar energy applications were not specifically designed for solar systems, as they were employed as imaging devices [74]. The imaging Fresnel lens operates by refracting light from an object and producing an image in the focal plane. However, due to imprecise manufacturing of the prism tips and grooves, these lenses suffer from aberrations. Despite this drawback, imaging Fresnel lenses are still commonly employed in applications such as monocrystalline photovoltaic electricity generation. Significant studies on the use of Fresnel lenses in concentrated solar energy applications for a variety of purposes can be grouped under two main fields: imaging Fresnel lens systems and non-imaging systems [75].

In Shanghai, China, Karimi et al. [76] designed a PV/T system concentrated with a linear Fresnel lens. The lens has a concentration ratio of 5.85X. It was found that the normal electrical efficiency decreased from 10.9 to 7.63% because of cloudy weather. However, under direct light, the power output significantly improved by 28%. The results also stated that the total thermal efficiency of the system improved in the range of 46.6–53%. Sornek et al. [77] presented a numerical, experimental study to improve the efficiency of a PV module using a Fresnel lens. The prototype was installed on the roof of the building in Krakow, Poland. Both the PV modules and the Fresnel lens faced south (the tilt angle of a PV module is 90° and the azimuth of the lens is 0°). The experimental setup contains two rows of PV modules with a Fresnel lens, In the first case, the Fresnel lens was located coaxially with the PV module, while in the second case, it was placed above the PV module, 75 cm higher. Authors concluded that Fresnel lenses allow integrated PV systems in buildings to operate more effectively and enhance the system performance by almost 7%.

Feng et al. [78] installed a concentrated PV/T system using a transmissive Fresnel lens as a solar concentrator to investigate and state the results of this system. The authors designed a model of a Fresnel lens (cycloidal transmissive) that can apply a particular focus line width. They created a wedge-shaped Fresnel lens and used 72-cell 5 W GaAs photovoltaic cells as the receiver which are thin film solar cells. An experimental study was performed under a cloudless sky and the experimental results showed that the highest electric efficiency is about 18% at noon, and the maximum heat receiving rate of cooling water is about 45%. The entire system showed an impressive performance with efficiency reaching 55%. Al Masud et al. [79] installed two different concentrated systems to compare parabolic trough and Fresnel lens concentrators to optimize the performance of PV and choose the best system. On a cloudless day, the results showed that the parabolic trough with an efficiency of 26.81% outperformed the Fresnel lens system with only 17.89% of output power. On a cloudy day, efficiency decreased slightly 22.3% and 14.1% respectively. Consequently, the authors determined that using a parabolic concentrating system is a better option than the Fresnel lens.

3.2. Medium concentration

The systems have medium concentration ratios in the range of 10X and 100X, and linear focusing systems like parabolic troughs or linear Fresnel lenses are commonly used. The heat produced by the cell is

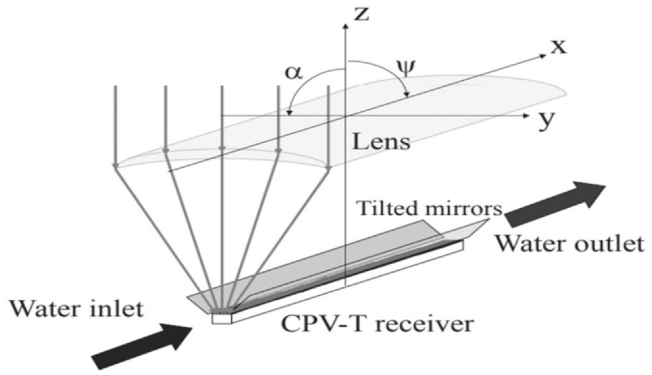


Fig. 10. Curved linear Fresnel lens installed on the dual-axis tracker [81].

these systems' biggest problem. The solar cell's temperature rapidly rises because of the concentration ratio, heat generated, and both. Many cooling methods are used to overcome the issues of heat dissipation; it is important to consider those efforts that increase the overall system cost. Sonneveld et al. [80] implemented an MCPV/T system by incorporating a planar Fresnel lens with a CR of 25x into the cladding of agricultural greenhouses. They constructed a 6 × 6 m greenhouse prototype with fixed linear Fresnel lenses and a PV/T collector. The PV/T module, featuring MCPV technology, was securely positioned using a dual-axis tracking system. The system instructed a thermal efficiency of 56% and an electric efficiency of 11%. In a separate study, Jaffré et al. [81] developed an MCPV-T system based on a curved linear Fresnel lens. The lens had a length of 1 m, a width of 0.46 m, and a theoretical focal length of 0.475 m. The system also had a theoretical CR of 20. The MCPV-T module was mounted on a 2-axis solar tracker, as illustrated in Fig. 10. The PV panel consisted of 36 small 20 × 20 mm monocrystalline silicon PV cells. The experimental results revealed the optical efficiency of the lens at 77% with a CR of 17, slightly below the theoretical value of 20. At a temperature of 50 °C, the cell efficiency was measured at 8.5%, while the thermal efficiency reached 50%.

In Australia, Coventry [82] Constructed a parabolic trough CPV/T system with a 1-axis tracking, 15 m long collector and geometric CR of 37x suns to examine the effects of non-uniform illumination on the solar cells. The collector consists of glass-on-metal mirrors that focus light onto high-efficiency monocrystalline silicon solar cells, while water flows through a conduit at the back of the panel to remove enduring energy. The proposed system demonstrated a thermal efficiency of approximately 58% and an electrical efficiency of about 11%.

3.3. High concentration

Such systems use either refractive or reflective technologies to obtain these high-concentration ratites which reach between 100 and 2000X. Linear concentration systems are used in a parabolic trough, or with linear Fresnel devices, such as lenses or mirrors. These systems are usually required for two-axis tracking. Developed high-concentration (HC) PV systems are used with high-efficiency III-V tandem solar cells. The cells have an extremely high conversion efficiency, and under concentrated light, the efficiency increases. Wang et al. [83] presented numerical and experimental study that investigate the electrical properties of multi-junction solar cells in HCPV systems. Based on a single diode, they created a numerical model for multi-junction solar cells. A high concentrated system with hyperboloid mirrors and a dual-axis tracker was installed in outdoor conditions. The results showed that the power of MJSC increased as the direct irradiation increased. With increasing the radiation from 650X to 800X, the results showed a significant improvement as the radiation intensity increased to 700 W/m^2, the result showed that the short circuit current increased from 1.002 to 1.243 A and open circuit voltage from 11.095 to 11.158 V. The maximum power increased from 9.466 to 11.358 W.

Xu et al. [84] developed a numerical simulation of a high concentrated PV/T system using Fresnel lenses as a solar concentrator and validated it with an experimental test. The HCPV/T module may achieve an electrical efficiency of 28% and a thermal efficiency of 60%, according to the data. Solar radiation has a greater impact on electrical efficiency than cell temperature. As irradiance increases, the thermal efficiency rises as well. As illustrated in Fig. 11, the CPV/T receiver is built and consists of a solar cell and an aluminium heat-sink. The solar cell is pasted at the front side centre of the heat-sink by thermal conductivity silica gel. On the back of the heat-sink, an axial grooved tube is manufactured, through which water runs and dissipates heat. Table 3 lists some of the previous advancements in concentration systems.

3.4. Technical challenges of (CPV) technology

Concentrating sunlight onto photovoltaic cells aims to maximize solar energy conversion. Yet, like all innovative technologies, CPV



Fig. 11. HCPV/T modules, [84]. HC = high-concentration; PV = photovoltaic.

Table 3
Summary of some previous studies for concentration systems

Reference	Study type	Work target	CPV type	Concentrator type	Main findings
Arshad et al. (2014) [65]	Experimental	Compare the efficiency of solar panels under different conditions.	LCPV	Flat Mirrors.	The mirrors and cooling outperformed the other configurations, achieving an efficiency of 52%.
Malik and Chandel (2020) [66]	Theoretical and Experimental	Enhance the electrical efficiency of PV by using Flat mirrors.	LCPV	Flat Mirror.	10–19.84% average power enhancement of the PV-reflector system in summers and 10–13.23% in winters.
Kolamroudi et al. (2022) [67]	Experimental	Increase solar reflection to enhance power output.	LCPV	Flat mirrors.	The system achieved a power output 2.84 times greater than traditional PV panels.
Abdullah et al. (2022) [68]	Experimental	Study the impact of utilizing reflective mirrors on the performance of the Photovoltaic/Trombe wall system.	LCPV	Flat Mirrors.	Maximum thermal efficiency reached 74.4%, and electrical efficiency 11%.
Lotfi et al. (2022) [69]	Experimental	Use metal fins and mirrors to enhance the irradiation received by the photovoltaic panels.	LCPV	Flat Mirrors.	The output power and electrical efficiency of PV panels improved by 11.4% and 13.1%, compared to the conventional case.
Hadavinia and Singh (2019) [70]	Theoretical and Experimental	Compare between compound parabolic concentrator and flat concentrator.	LCPV	2D Symmetric CPC.	The CPC concentrator outperforms the V-trough by 2.4% in power output.
Yousef et al. (2016) [71]	Theoretical and Experimental	Compare between CPC -PV and flat PV modules with and without cooling.	LCPV	2D Symmetric CPC.	CPC-PV module outperforms the flat PV module by 33% in power generation. with cooling, PV and CPC-PV modules produced 11% and 15% more power.
Li et al. (2020) [72]	Theoretical and Experimental	Performance comparison of two CPC-PV systems which having identical CR but different geometric scales.	LCPV	2D Symmetric CPC.	The small-scale CPC-PV system had higher maximum electrical power output and conversion efficiency.
Jaaz et al. (2017) [73]	Experimental	PV/T Performance analysis using CPC and jet impingement.	LCPV	CPC mirrors.	The electrical efficiency was improved by 7% when using CPC and jet impingement cooling, and the power output improved by 36% compared to using jet impingement cooling alone.
Karimi et al. (2017) [76]	Experimental	Analyze experimental results of solar thermal and solar PV/T systems using Fresnel lens.	LCPV	Linear Fresnel lens.	The electrical efficiency decreased to 7.63%, The output power of the PV module increased by 28%. The total efficiency of PV/T systems under linear solar focusing reached 53%.
Sornek et al. (2018) [77]	Theoretical and Experimental	Improve the efficiency of a PV module using a Fresnel lens.	LCPV	Linear Fresnel lens.	The performance of the PV module increased by about 7%.
Feng et al. (2016) [78]	Theoretical and Experimental	Analyze of CPV/T system using Fresnel lens with cycloidal outside surface.	LCPV	Cycloidal Fresnel lens.	The Maximum power efficiency is about 18%. The maximum thermal efficiency is about 45%. The total efficiency can reach more than 55%.
Al Masud et al. (2022) [79]	Experimental	Compare parabolic trough and Fresnel lens concentrators to optimize the best performance of PV.	LCPV	Parabolic trough and Fresnel lens	The solar panel's output power improvement is 17.89% for the Fresnel mirror concentrator and 26.81% for the parabolic trough. Both concentrators increase output power by 14.1% and 22.3%.
Sonneveld et al. (2011) [80]	Experimental	Evaluate the performance and feasibility of a concentrated PV/T system utilizing linear Fresnel lenses in a greenhouse setting.	MCPV	Planar Fresnel lens	The PV/T module showed a thermal efficiency of 56% and an electric efficiency of 11%.
Jaffré et al. (2014) [81]	Experimental	Develope and evaluated a new curved linear Fresnel lens concentrating PV and thermal system for enhanced solar energy utilization and efficiency.	MCPV	Curved linear Fresnel lens	The optical efficiency of the lens was 77%. The cell efficiency was 8.5%. Thermal efficiency was 50%.
Coventry (2005) [82]	Experimental	Investigate the performance of a parabolic trough PV/T collector, focusing on the impact of non-uniform illumination on the solar cells.	MCPV	Parabolic trough	Thermal efficiency is around 58% and electrical efficiency is around 11%,
Wang et al. (2022) [83]	Theoretical and Experimental	Analyze the electrical properties of multijunction solar cells using HCPVs with hyperboloid mirrors.	HCPV	Hyperboloid mirrors.	Radiation intensity increased to 700 W/m$_2$. The maximum power increased from 9.466 W to 11.358 W.
Xu et al. (2016) [84]	Theoretical and Experimental	Analyze the electrical and thermal performance of a HCPV/T module equipped with a point-focus Fresnel lens.	HCPV	Fresnel lens.	The HCPV/T module achieve an electrical efficiency of 28% and a thermal efficiency of 60%.

CPCs = compound parabolic concentrators; CPV = concentrated photovoltaic; HC = high-concentration; LC = low concentration; MC = medium concentration; PV = photovoltaic; PV/T = photovoltaic thermal.

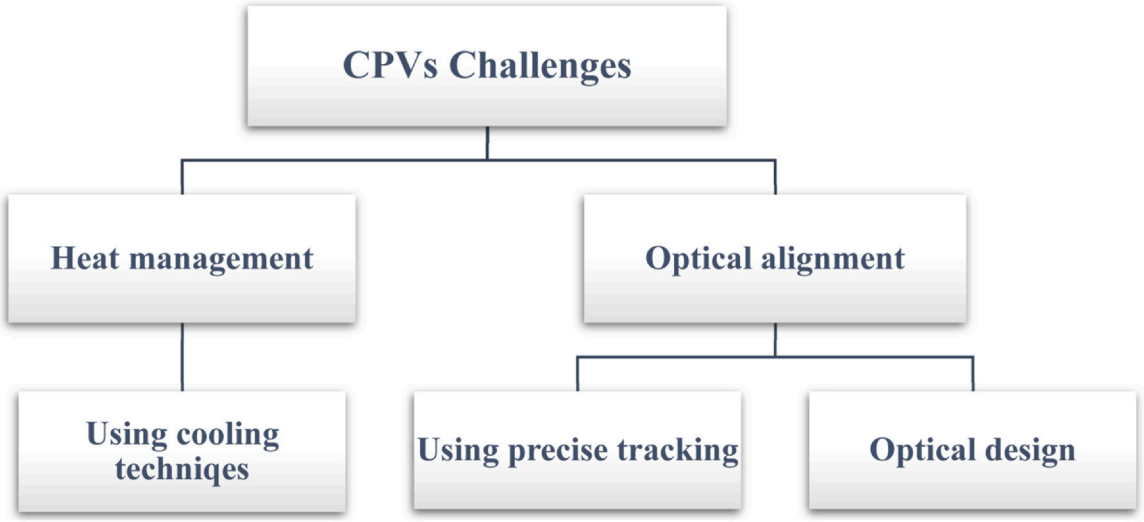


Fig. 12. CPVs Challenges and solutions. CPV = concentrated photovoltaic.

systems faces a series of technical challenges that must be effectively managed to ensure practical deployment and long-term viability. In this section we examine the key technical challenges encountered in CPVs and explores the innovative solutions that have been developed to overcome these challenges. The challenges and solutions are illustrated in Fig. 12.

3.4.1. Heat management challenges

Optical devices concentrate solar irradiance onto a small area of solar cells, increasing it by up to a thousand times. This high level of solar radiation raises the temperature of the PV cell, leading to a decrease in its efficiency and lifespan. So efficient thermal design (cooling techniques) is necessary to solve these challenges [85].

Here's some brief studies of the cooling techniques actually used with the concentration system to minimize the temperature of CPV module to solve heat management issues.

Saadah et al. [86] investigated the impact of graphene fillers on the performance of Thermal Interface Materials with the (CPV) solar cells and conducted tests to cool solar cells using optical devices with a concentration ratio up to 2000. Researchers found that using graphene recovered a significant amount of energy loss due to the high temperature of solar cells. Wang et al. [87] performed a numerical and experimental analysis to evaluate the impact of passive cooling and active cooling on CPV hybrid solar system, the results showed that active air cooling is more efficient than passive air cooling, it's found that active air cooling system maintain PV panel temperature below 75 ˚C, when the atmosphere temperature was 45 ˚C. Deymi-Dashtebayaz et al. [88] conducted a novel CPV/T system with Al_2O_3 nanofluid flow into porous channel coupled with a domestic heat pump (CPV/T-HP), the result showed that the PV cell efficiency increased by 3.5% and the panel surface temperature decreased by an average 3 K with increasing the mass flow rate of the nanofluid from 0.01 to 0.04 kg/s.

Tang et al. [89] developed manifold ultrathin pin fin heat sink for cooling a CPV/Ts under a high solar concentration of 1000 suns, it's found that the CPV panel temperature can be maintained 51 ˚C at 3.5 kg/h wind velocity and 25 ˚C ambient temperature. The results also revealed that increasing the flow rate enhances electrical efficiency.

3.4.2. Optical alignment challenges

The Optical alignment is the technical difficulties and complexities associated with achieving and maintaining precise orientation and positioning of optical components such as mirrors, lenses. These challenges are critical because precise optical alignment directly impacts the efficiency and performance of CPV systems in capturing solar energy and converting it to electricity. Therefore, Optical design is essential to minimize reflection losses and maximize light concentration. Also achieving precise solar tracking accuracy is essential to maintaining optimal alignment with the sun's position throughout the day. Inaccurate tracking can lead to suboptimal power capture [85].

Here's some previous studies related to tracking system and optical design.

Satué et al. [90] conducted a study comparing five different control strategies for HCPV sun tracking. the research aimed to analyze and optimize the energy production efficiency of (HCPV) systems by evaluating various control methods, including open-loop strategies using Solar Equations and closed-loop strategies utilizing feedback from electro-optical sensors and DC power sensors. It's found that the measured loss in efficiency when the sun-tracker is not precisely calibrated is about 6%. This highlights the importance of accurate tracking for maximizing energy output in (HCPV) modules.

Yazdanbakhsh et al. [91] conducted a study to investigate the efficiency of a (CPC) equipped with a tracking system in the solar photodegradation of carbamazepine (refractory organic compound, from aqueous solutions), results showed that using a solar tracking system with (CPC) can enhance the photo-thermal conversion efficiency by 2–3 times during the degradation of organic compounds. Miñano et al. [92] Stated that moving forward, less effective and cheaper optical designs are viewed as significant obstacles in CPV technology, hindering its widespread adoption. Several concentrators like FK and F-RXI have been suggested as potential game-changers in the CPV field, although they still need to be tested.

Chong et al. [93] discussed the importance of optical design in (CPVs), emphasizing progress in high-efficiency solar cells and optical concentrators. They evaluate different optical systems suggested by professionals, highlighting the importance of affordable and efficient solutions. The research looks into various solar concentrator types and optical configurations, focusing on overcoming durability and reliability issues in CPVs. In general, the study seeks to improve the effectiveness and affordability of generating solar energy by using creative optical design solutions.

4. Spectral splitting filtering

Solar cells own the potential to convert a specific wavelength range of sunlight into electricity, as the effective spectrum they can harness is limited to the full spectral range of solar radiation [94]. When exposed to rays outside this range, the solar panel's temperature increases, resulting in decreased electrical conversion efficiency. Traditional cooling

technologies rely on extracting heat from the panel in so-called "post-absorption" heat management solutions. While the spectral splitting technique is considered another solution to address the heating problem, which is to detect the specific spectral range in which the cells have the best efficiency and prevent the unwanted part before reaching the panel in what is known as the "pre-absorption" method. The spectral splitting technique includes dividing the incident rays on the panel into two separate parts. The first part consists of the requested wavelength range that PV panels can efficiently convert into electricity. The second part, including wavelengths outside the requested range, leads to an increase in panel temperature. This excess heat can be employed for other applications or directed towards another panel designed for different wavelength ranges. This technique can be accomplished by using different optical components and employing two primary methods. The first method involves the use of a thin-film wave interference filter, known as the reflective method. The second, known as the absorptive method, utilizes a selective absorber. A comprehensive depiction of these two methods is illustrated in Fig. 13.

4.1. Reflective method

The approach applied in this method involves the deposition of multiple layers of non-absorbing dielectric materials with varying refractive indices onto a transparent substrate. Also, it can be applied by using a transparent layer with a continuously changing refractive index to achieve the desired outcome. This arrangement effectively splits the incident's light into two distinct parts, each travelling in a different direction. Qiu et al. [96] developed a hybrid CPV/T spectral splitting system based on a wave-selecting filter-coated CPC and linear Fresnel reflector. This allows visible and near infrared spectrum to transmit onto the PV panel, while the rest are reflected to a receiver tube. This achieved an exergy efficiency of 35.51% and an optical efficiency of 94.78%. Crisostomo et al. [97] developed an analytical optimization method to determine optimal wavelengths for solar spectrum division by utilizing SiN_x/SiO_2 (high and low refractive index materials) multilayer thin film filters for spectrally selective beam splitting and a linear Fresnel lengths (low concentration) PV/T collector. In addition to, Ray tracing software model (Zemax 12) to allow for estimation of collector optical efficiency at various sunlight angles of incidence. The result revealed a 47% increase in power delivery compared to a Si CPV standalone system under the same concentration ratio. Mojiri et al. [98] designed a hybrid solar receiver that integrates a simple dichroic filter with absorbing liquid channel. The configuration consists of dichroic coating (acted as a long pass filter) that was designed to direct specific wavelengths to (PV) cell, while the liquid channel (acted as a short pass filter) absorbed the remaining wavelengths for heat conversion. The modeling findings indicate that receiver can redirect 54.5% of solar spectrum to PV cells. Around 73.3% of this energy falls within the 600–1125 nm band. The liquid channel absorbs the rest of the solar spectrum, converting it to heat. Kandil et al. [99] conducted both theoretical and experimental study on a PV hybrid system. The system consisted of a Fresnel lens, a cold mirror as a spectral splitting filter, a cooling system with PCM and a 2-D tracking system, as depicted in Fig. 14. The authors identified a parameter entitled the utilization factor, which is the ratio between actual energy output and maximum energy output that could be obtained. It was found that utilizing the beam splitter increased the system's net power by about 43% compared to the stand-alone PV system. The study also revealed that the utilization factor for conventional PV remained at 0.13, while for this proposed system, the factor demonstrated an enhancing trend of 0.31, 0.41, and 0.46 with increasing the solar radiation levels 900, 950, and 1000 W/m^2.

Huaxu et al. [100] designed a CPV/T system utilizing an optical splitting film as a spectral splitting system. The main objective of the configuration was to employ the entire spectrum of sun rays while at the same time reducing the temperature of the CPV cell. To achieve this objective, a SiO_2/TiO_2 thin film layer was used, exposing reflection bands that aligned well with the spectral response bands of the CPV cell. It was found that this alignment disallowed infrared rays from heating up the CPV cell, and generally enhancing the overall system performance. This configuration resulted in an 11 °C reduction in CPV cell temperature, a 9.4% increase in power generation, and a 1.3% enhancement in photoelectric efficiency. The overall energy and efficiency of the system reached 18.85% and 14.61%, respectively, which were 5.8% and 1.5% higher than CPV without film concentration. Li et al. [101] proposed a non-confocal, two-stage parabolic dish-CPV/T hybrid system that applies spectral beam splitting technology to effectively employ full-spectrum solar energy. The system attained an optical efficiency of 80.7% and a stable energy efficiency above 20.9%. The inauguration of a spectral beam splitter increased the efficiency of the PV/PT component and the overall system efficiency by about 17.6% and 10.2%, respectively.

Alnajideen and Min [102] presented a PV panel integrated with a thermoelectric (TE) and employed dichroic mirrors to enhance overall efficiency. The concentrator reflected and concentrated the visible spectrum onto a PV cell while allowing the infrared spectrum to pass through to a thermal absorber for thermoelectric generation. The overall efficiency of the integrated PV–TE system reached 16.9%, representing a 6.3% increase compared to the use of spectral aluminium mirrors. Moreover, the power output increased from 12.41 to 62.04 mW. Ling et al. [103] proposed a spectral-splitting PV thermochemical system, which involved preheating liquid methanol and solar

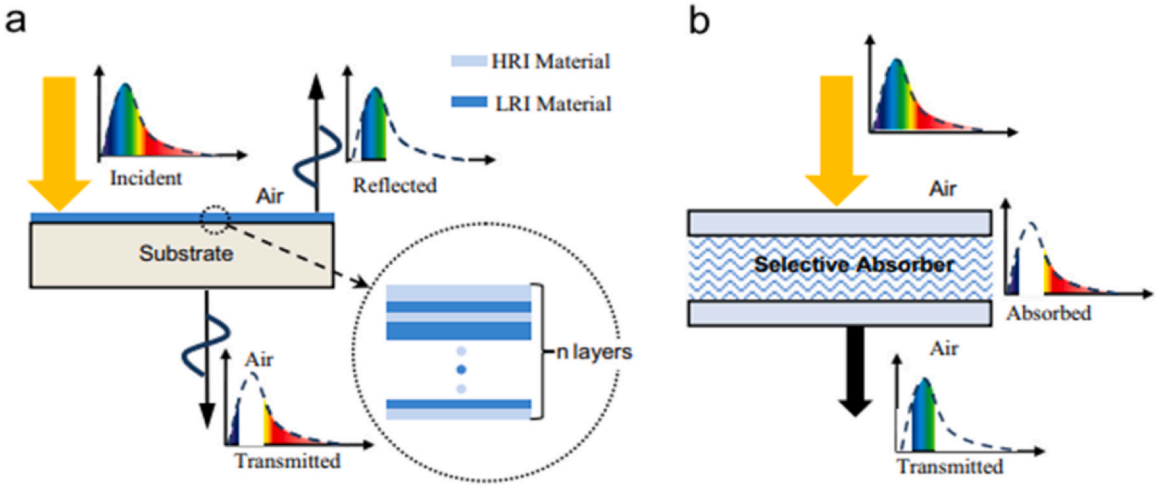


Fig. 13. Spectral splitting methods: (a) Reflective method. (b) Absorptive method [95].

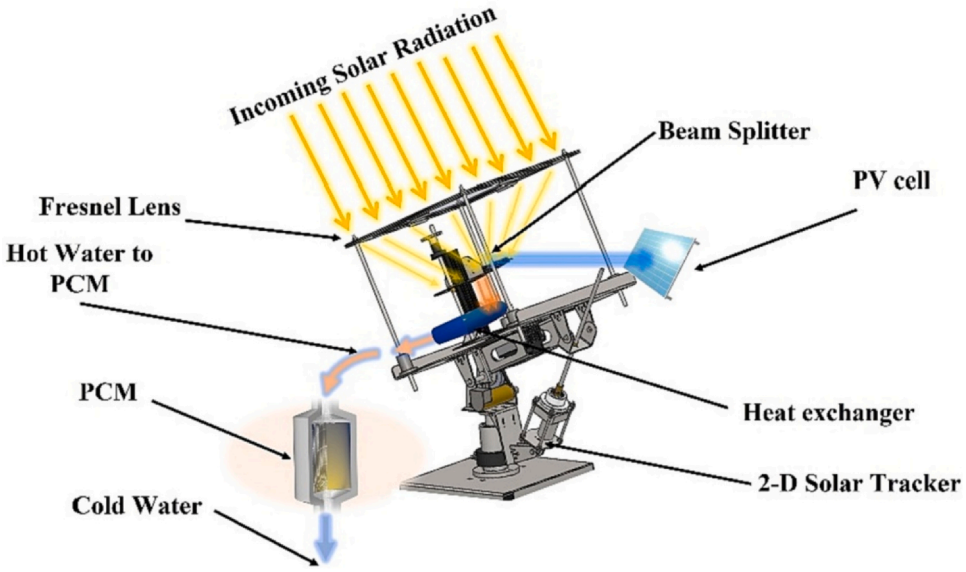


Fig. 14. Design of the hybrid system [99].

power generation. This system achieved an efficiency of 39% by decoupling the temperature between photovoltaics and thermochemistry through spectral filtering. The study also highlighted the cost reduction potential and advancements in PV/T technologies achieved by utilizing Fresnel reflectors. Additionally, solar thermal energy storage efficiency exceeded 60%.

4.2. Absorptive method

In this method, selective materials with specific optical properties that have the ability to absorb a certain wavelength of incident light and transmit the rest are used to prevent the unwanted range from reaching the PV panel. In a study conducted by Huang and Markides, [104] three types of semi-transparent PV were utilized, namely cadmium telluride (CdTe), perovskite (PVSC), and polymer solar cells (PSC), to investigate their effectiveness in spectral splitting and electricity generation. The results revealed that the conversion efficiency of solar energy to electricity varied across the different panel types. Specifically, the CdTe panel demonstrated an electricity conversion rate of 8.6%, while the PVSC panel achieved 14.2%, and the PSC panel attained 5.5%. Moreover, the study examined the heat generation within the solar cells for each panel type, indicating percentages of 55.4%, 41.8%, and 32.2% for CdTe, PVSC, and PSC panels respectively. Additionally, the analysis evaluated the solar energy absorption by the solar thermal absorber, resulting in absorption percentages of 27.2%, 33.3%, and 51.8% for CdTe, PVSC, and PSC panels respectively.

Han et al. [105] investigated the effect of spectral splitting on a CPV/T system using combined optical filters, including a quartz container with HB650 filters filled with propylene glycol (PG). The use of optical filters decreased the conversion of solar energy into electricity, but it increased the overall efficiency. Specifically, the results showed that the solar energy converted to electricity was 12.73%, 10.15%, and 6.03% for the system without a filter, with a PG filter, and with a PG + HB650 combined filter, respectively. The total efficiency was 23.88% and 46.77% for the system with a PG filter and a PG + HB650 combined filter, respectively. Elharoun et al. [106] present an experimental and numerical study that examined the optical filter thickness effect and the concentration of nanoparticles on the performance of a PV panel. The panel is linked with a parabolic type of concentrator and uses a ZnO-water nanofluid as a coolant fluid. The experiments involved different thicknesses (1 cm, 1.5 cm, and 2 cm) and nanofluid concentrations (50 ppm, 100 ppm, 150 ppm, and 200 ppm). It was found that by increasing the filter thickness, electrical efficiency increased from 6.2% for 1 cm thickness to about 7.9% for 2 cm thickness. Also, thermal efficiency increased from 21.5 to 41.3%. The results also showed that electrical efficiency increased from 6.2 to 7.7% by increasing nanoparticle concentration up to 100 ppm, but then decreased at 200 ppm to about 6.2%. While thermal efficiency increased from 21.5 to 31.4% by increasing the concentration.

Elharoun et al. [107] investigated the effect of the air gap between the PV panel and the optical filter (ZnO-water nanofluid flows between two layers of acrylic). The experiments involved comparing different air gap distances (1 cm, 3 cm, and 5 cm) for bare PV, no-filter CPC-PV and filtered CPC-PV configurations. The study found that the 3 cm air gap resulted in the highest electrical, thermal, and overall efficiency, which were 17.1%, 21%, and 32.8%, respectively. Additionally, a comparison between horizontal and inclined (30° tilt angle) panels revealed higher efficiencies for the inclined configuration. Xia et al. [108] investigated a PV/T system with spectral splitting technology using a mixture of carbon nanotubes (CNT) and Ag nanofluids. The study examined the spectral properties of Ag, CNT, and CNT/Ag nanofluids at various concentrations. Experimental results showed that at a concentration of $5 \times 106\,\mu g/m^3$, the electrical and thermal efficiency of the PV/T system reached 8.2% and 45%, respectively.

Meraje et al. [109] investigated experimentally a Fresnel concentrated PV/T system integrated with a nanofluid spectral splitter. The system employed a linear Fresnel lens to concentrate solar irradiation on a series of tubes through which the nanofluid streamed. The surface temperature of the PV module remained significantly lower than the temperatures of the nanofluid and outflow water. By varying the concentration of ZnO in the nanofluid, the optimum total efficiency was found to be 85% at a concentration of 0.0089%. An initial experiment conducted by An et al. [110] explored the application of a solar distillation system using a PV/T system based on the nano-spectral splitting method. To capture solar light that conventional PV cells cannot utilize, gold (Au) and Ag nanoparticles were introduced into water, enabling its conversion into heat. A comparison with the standard system revealed that the enhanced systems, incorporating Au and Ag nanoparticles at a concentration of only 26 ppm, exhibited a significant increase in water output of 79.9% and 69.4%, respectively. Furthermore, the thermal efficiency of the gold and silver nanoparticle-enhanced systems experienced improvements of 6.8% and 4.8%, respectively.

In another study conducted by An et al. [111], an experiment was carried out utilizing a Cu_9S_5-octylamine nanofluid spectral splitting filter to overcome the limitations of traditional PV/T systems. The

findings demonstrated that the PV/T collector equipped with the Cu9S5 nanofluid filter achieved a remarkable maximum overall efficiency of 34.2%, surpassing the efficiency without a filter by 17.9%. Moreover, this system proved capable of attaining thermal energy at temperatures exceeding 100 °C. Incorporating a glass cover resulted in a marginal reduction of 1.1% in the PV efficiency of the collector; however, it contributed to an increase in both thermal efficiency and overall efficiency by 4.2% and 3.1%, respectively, compared to a system without a glass cover. He et al. [112] conducted research on using an Ag@TiO_2/ethylene glycol/water nanofluid as a beam splitter for PV/Ts applications in cold regions. They aimed to enhance the performance of solar arrays in cold climates. Their results showed that the nanofluid-based beam splitter significantly increased the energy conversion efficiency of the system, making it a promising choice for PV/T applications in cold regions as it recorded 83.7% overall solar energy utilization efficiency (concentration of nanoparticle was 200 ppm). Which was 1.5-fold higher performance compared to ethylene glycol/water solution, and 4.4-fold higher performance compared to silicon cell alone. Furthermore, An et al. [113] conducted an experiment to assess the performance of a spectral splitting PV/T hybrid system employing a polypyrrole nanofluid, as depicted in Fig. 15. The results indicated that the hybrid PV/T system with the polypyrrole nanofluid filter achieved an impressive overall efficiency of 25.2%, surpassing the system without the filter by 13.3%. Although the PV efficiency of the system with the polypyrrole nanofluid filter was slightly lower than that of the system without the filter, the system's performance remained acceptable, particularly considering the attainment of medium-temperature thermal energy.

Lastly, Huang et al. [114] developed a comprehensive two-dimensional model of a fluid-based spectral splitting PV/T collector by incorporating a filter flow channel to enhance its electrical and thermal performance. Employing Si400–1100 as the optical filter, the system achieved high-temperature heat generation of 400 °C with an efficiency of 19.5%, low-temperature heat production of 70 °C with an efficiency of 49.5%, and electricity generation with an efficiency of 17.5%. Table 4 reviews some of the studies that deliberated spectral splitting filtering systems.

5. Tracking systems

Solar tracking systems are advanced technologies that have been designed to optimize solar energy collection. The concept of these systems is based on continuously tracking the sun's movement during the day, guaranteeing that solar panels capture the maximum amount of sun rays. Solar panels face the problem of partial shading, the effect of which is usually reduced through different types of connections for the arrays of panels [115], but tracking contributes greatly to reduce this problem by detecting the presence of an obstacle blocking the rays and then changing the position of the panel so that it remains exposed to the rays. As a result, increasing the energy output from the panel and enhancing the overall performance. Fig. 16 provides an overview classification of the various tracking methods.

5.1. Driving system

To follow the sun's path across the sky, a driving system of a solar tracker must be applied to the configuration. By combining this system with solar panels, trackers allow these panels to self-align with the sun, which leads to higher renewable energy generation. These driving systems can be classified into three types: passive tracking, active tracking, and semi-passive tracking.

5.1.1. Passive tracking

Passive tracking operates by employing thermal expansion phenomena in materials or using the imbalance in pressure between two points at opposite ends of the tracker. In general, a fluid whatever liquid or gas is used inside two reservoirs placed opposite to each other. The reservoirs design permits the vaporization of the fluid, adjusting its characteristics in response to the sun's path changing over the daytime. The system is activated through a link between the two reservoirs, allowing the condensate fluid to flow from the reservoir with high solar radiation to the small [116]. So in general, such systems depend on compressed fluid with a low boiling point, which generates an imbalance and produces movement in the tracker structure [116]. The passive systems are less precise than active but could influenced by wind or temperature fluctuations. Furthermore, at specific times of the year, manual adjustments may be necessary. Unlike active trackers, passive tracking does not use mechanical or electrical components such as motors or sensors in its track mechanism.

Clifford and Eastwood [117] developed an automated passive solar tracker that autonomously monitors the sun's movements. It was found that this system demonstrated an efficiency increase of up to 23% compared to fixed solar panels. Natarajan and Srinivas [118] conducted a mathematical and experimental study to define the optimal tracking wheel radius for a continuous, power-free tracking system. The tracking system was connected to a linear solar concentrator. The proposed system established an average enhancement of about 1.5% at the global component and 3% at the beam component compared to the intermittent tracking approach. Alemayehu and Admasu [119] proposed a passive solar tracker using a bimetallic strip deflector that produces an additional power of 47 W per unit area, with a daily average power output range between 70 and 120 W. The results showed that the energy collecting efficiency increased by 24.86% and the accuracy by 96.4% compared to an ideal tracking system, while it costs half as much as a traditional solar tracker.

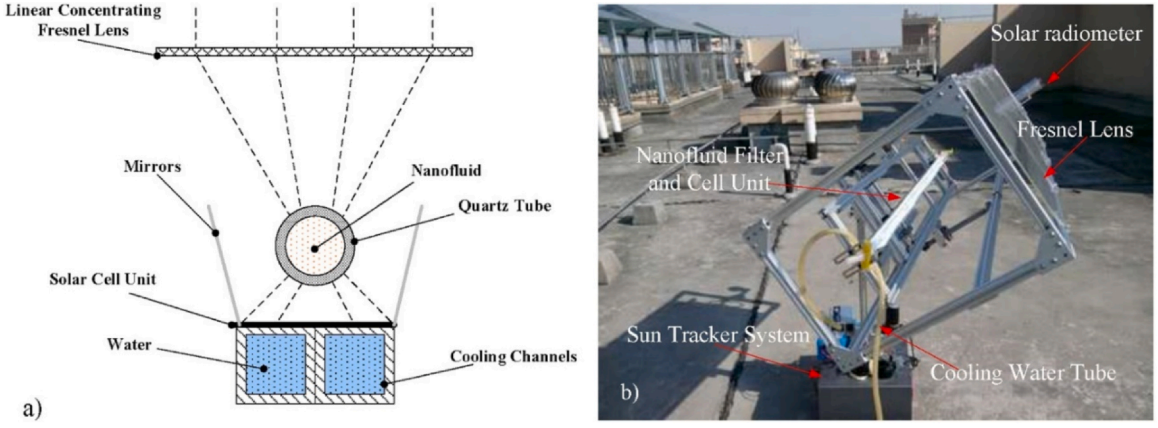


Fig. 15. Spectral splitting PV/T hybrid system: (a) configuration sketch; (b) system picture [113]. PVT = photovoltaic thermal.

Table 4
A comparative study of spectral splitting filtering systems

Reference	Study type	Work target	Filter type	Main findings
Qiu et al. (2024) [96]	Theoretical and Experimental	Develope a hybrid CPV/T spectral splitting system based on a wave-selecting filter-coated CPC and linear Fresnel reflector.	Wave selecting filter-coated CPC and Fresnel reflector.	This achieved an exergy efficiency of 35.51% and an optical efficiency of 94.78%.
Crisostomo et al. (2015) [97]	Theoretical	Develope an analytical optimization method to determine optimal wavelengths for solar spectrum division by utilizing SiN_x/SiO_2 multilayer thin film filters and linear Fresnel mirrors.	SiNx and SiO_2 multilayer thin film filter	The result revealed a 47% increase in power delivery compared to a Si CPV stand-alone system under the same concentration ratio
Mojiri et al. (2015) [98]	Theoretical and Experimental	Design a hybrid solar receiver that integrates a simple dichroic filter with absorbing liquid channel.	Simple dichroic filter with absorbing liquid channel	Receiver can redirect 54.5% of solar spectrum to PV cells. 73.3% of this energy falls within the 600–1125 nm band. The liquid channel absorbs the rest of the solar spectrum, converting it to heat.
Kandil et al. (2024) [99]	Theoretical and Experimental	Investigate the performance of a hybrid system consisting of PV/PCM with spectral splitting, and a 2-D tracking system.	Cold Mirror	With increasing solar radiation (900, 950, and 1000 W/m^2), the utilization factor for the traditional PV remained constant at 0.13, while for this hybrid system, it tends to increase (0.31, 0.41, and 0.46).
Huaxu et al. (2020) [100]	Experimental	Devise an optical splitting film to take full benefit from the sunlight and reduce the cell temperature.	Reflection bands of TiO_2/SiO_2	- Cell temperature was reduced by 11 °k. - Increased power generation by 9.4%, photoelectric efficiency by 1.3%, and the overall efficiency of the system by 1.5%, compared to CPV without film concentration.
Li et al. (2023) [101]	Theoretical	Propose a hybrid PV/T system with non-confocal parabolic dish.	A glass substrate covered with a stack of optical films composed of a series of material arrangements with different refractive indexes.	- The energy efficiency is stable above 20.9%. - Introducing the spectral beam splitter increased the photovoltaic/photothermal component efficiency and system exergy efficiency by 17.6% and 10.2%.
Alnajideen and Min (2022) [102]	Experimental	Propose a hybrid PV-TE system to improve overall efficiency.	Dichroic mirrors	- The overall efficiency of the hybrid PV–TE system reaches 16.9%, which increased by 6.3% compared to the use of spectral aluminium mirrors.
Ling et al. (2020) [103]	Theoretical	Propose a spectral splitting PV thermos chemical system for energy storage by preheating liquid methanol, and solar power generation.	Fresnel reflectors with Specific mirrors (70 layers of Ta_2O_5/SiO_2 coating).	Achieved a net solar electric efficiency of about 39%, and solar thermal energy storage efficiency of about 60%.
Huang and Markides (2021) [104]	Theoretical	Compare three types of semi-transparent panels.	CdTe, PVSC, and PSC panels.	- PVCS has the best electrical efficiency at 14.2%. - CdTe has the best thermal efficiency at 55.4%.
Han et al. (2021) [105]	Experimental	Test the effect of spectral splitting on the CPV/T system by using a combined optical filter in a quartz container.	HB650 filters, filled with propylene glycol.	Total efficiency is 23.88% for the system with PG filter, and 46.77% for PG + HB650 combined filter.
Elharoun et al. (2023) [106]	Theoretical and Experimental	Test the effect of the optical filter thickness and nanoparticle concentration on a PV panel with a parabolic concentrator.	ZnO-water nanofluid flows between two layers of acrylic.	100 ppm is the optimum concentration with an electrical efficiency of 7.7%, while the thermal efficiency increased continuously from 21.5% (at 50 ppm) to 31.4% (at 200 ppm).
Elharoun et al. (2023) [107]	Experimental	Test the effect of changing the air gap between the PV panel and the optical filter.	ZnO-water nanofluid flows between two layers of acrylic.	The 3-cm air gap has the highest electrical, thermal, and overall efficiency, which were 17.1, 21, and 32.8% respectively.
Xia et al. (2023) [108]	Theoretical and Experimental	Investigate the spectral properties of Ag, CNT, and CNT/Ag nanofluids at various concentrations.	Carbon nanotubes and silver nanofluids.	The filtration efficiency has the highest value at the concentration of 5×10^6 μg/m^3, with electrical and thermal efficiency of 8.2% and 45%, respectively.
Meraje et al. (2022) [109]	Theoretical and Experimental	Use Fresnel concentrated PV/T system with different concentrations of nanofluid as a spectral splitter.	Series of tubes through which ZnO nanofluid flows.	The optimum total efficiency was 85% at a concentration of 0.0089%.
An et al. (2018) [110]	Experimental	solar distillation system using PV/T system based on nano-spectral splitting method with 26 ppm concentration.	Au and Ag nanofluids.	For gold and silver nanoparticles, the power output increased by 79.9% and 69.4%, respectively. the thermal efficiency increased by 6.8% and 4.8%, respectively compared to standard system.
An et al. (2016) [111]	Experimental	Examine the performance of a CPV/T collector with a spectral splitting filter.	Oleylamine solution of Cu_9S_5 nanoparticle filled in a quartz tube.	The system achieved a maximum overall efficiency of 34.2%, which was 17.9% higher than the efficiency without a filter.

(*continued on next page*)

Table 4 (*continued*)

Reference	Study type	Work target	Filter type	Main findings
He et al. (2019) [112]	Theoretical	Use Ag@TiO_2/ethylene glycol/water nanofluid as a beam splitter for PV/Ts applications in cold regions.	Ag@TiO_2/ethylene glycol/water nanofluid	Recorded 83.7% overall solar energy utilization efficiency (concentration of nanoparticle was 200 ppm). Which was 1.5-fold higher performance compared to ethylene glycol/water solution, and 4.4-fold higher performance compared to silicon cell alone.
An et al. (2016) [113]	Experimental	Conduct an experiment to investigate the performance of spectral splitting on a PV/T hybrid system.	Polypyrrole nanofluid.	The system achieved an overall efficiency of 25.2%, which is 13.3% greater than the system without the filter.
Huang et al. (2021) [114]	Theoretical	Develope a 2D model of a fluid-based spectral splitting PV/T collector by adding a filter flow channel.	Si400–1100	The system generated high-temperature heat of 400 °C with an efficiency of 19.5%, low- temperature heat of 70 °C with an efficiency of 49.5% and electricity with an efficiency of 17.5%.

CNT = Carbon nano tube; CPCs = compound parabolic concentrators; CPV = concentrated photovoltaic; PCM = phase change material; PSC = polymer solar cell; PV = photovoltaic; PVSC = perovskite solar cell; PV/T = photovoltaic thermal.

5.1.2. Active tracking

An active solar tracking system is a complex mechanism that uses some sensors to determine the sun's path through the day perfectly. This type of tracking system includes mechanical parts such as motors and sensors to adjust the position of the drive system exactly related to the sun's movement [18]. Some of these systems can adjust solar panels both horizontally and vertically for optimum alignment with the sun. Solar sensors are located beside or on the collectors or panels and monitor the sun's position continuously regarding the solar panels or the collectors. Riyadi et al. [120] conducted a study involving three panels: fixed, 1-axis, and 2-axis solar tracking. They employed light-dependent resistors, a microcontroller, and thermocouples, and measured the power output. The recorded energy outputs were 5.6, 6.8, and 7.5 kWh/m^2/d. The maximum power increased by 0.77, 1.19, and 1.65 W, with 84% originating from PV power and 16% from the thermoelectric generator. The maximum efficiency improved by 3% compared to the stand-alone system, reaching values of 10.57%, 12.53%, and 13.99%.

5.1.2.1. Open loop. Also known as open cycle, the open-loop solar tracking system adjusts the orientation of solar panels or PV arrays based on predetermined algorithms or time schedules, without continuously monitoring the sun's position or environmental conditions. Electric or hydraulic motors are incorporated into open loop tracking systems to move the solar panels or arrays, while a microcontroller or computer calculates the appropriate positioning angles based on the time of day, date, and geographic location. Open-loop tracking systems are often simpler and more cost-effective than closed-loop tracking systems, and they may have lower maintenance requirements due to the fewer components and sensors they utilize. However, they rely on calculated data, which may not always accurately reflect real-world conditions. Factors such as cloud cover, atmospheric conditions, and unexpected deviations from predicted or calculated solar positions can impact their accuracy. Pirayawaraporn et al. [121] developed an innovative sensor-less tracking system to address the limitations of existing sensor-less solar tracking systems by using a particle filter, and sun position sensors. The results show that the system enhances energy generation performance by 20.1% compared to the fixed plate system. Mi et al. [122] developed an open-loop system that allows for active control and adjustment of the tracking frequency, resulting in improved tracking efficiency compared to a constant tracking frequency. By establishing a correlation between the azimuth and altitude angles, the system achieves energy savings and improved stability. Fathabadi [123] introduced a novel sensor-less dual-axis solar tracking system that achieves high accuracy. It's controlled by a maximum power point tracking unit. This system boosts energy efficiency by 28.8–43.6%.

5.1.2.2. Closed loop. Also referred to as a closed-cycle system, a type of solar panel tracking system actively monitors the sun's position and continuously adjusts the orientation of solar panels or PV arrays to maximize energy capture through a feedback mechanism. The closed-loop tracking system incorporates a control algorithm that processes data from sensors and calculates optimal angles for the solar panels based on real-time information. Actuators such as electric motors, hydraulic systems, or other mechanical devices adjust the orientation of the solar panels [18]. The system employs sun tracking sensors like light-dependent resistor (LDR) sensors to detect the sun's position, solar irradiance sensors to measure sunlight intensity, and environmental sensors to monitor temperature, wind, and other factors. Due to continuous monitoring and real-time adjustments, these systems offer higher accuracy compared to fixed solar panels or open-loop tracking systems.

Jamroen et al. [124] developed and implemented an automatic dual-axis solar tracker system by using LDR sensors. The system increased electrical energy efficiency by 44.89% compared to a fixed flat-

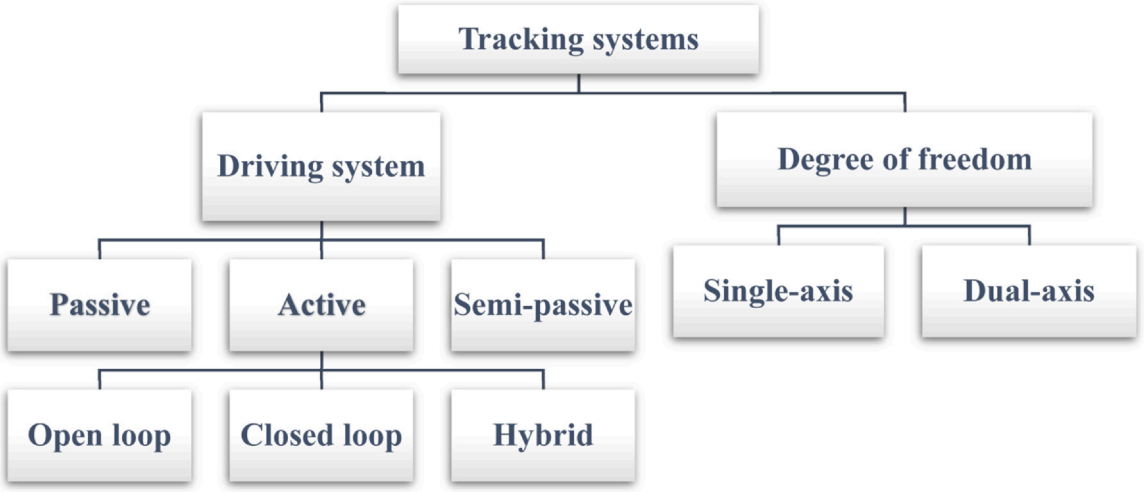


Fig. 16. Tracking systems classifications.

plate system. Saldivar-Aguilera et al. [125] introduced a dual closed-loop control strategy for single-axis solar trackers and did experimental comparisons between a typical single and dual control loop. The results showed a 78% reduction in mean solar tracking error with the suggested approach as the error value reaches 0.21° compared to the 0.97° error obtained with the simple control. The proposed system offers a viable and cost-effective alternative, showcasing greater stability and robustness in solar tracking. Garrido and Díaz [126] present a cascade control algorithm designed specifically for HCPV systems that minimizes the sun tracking error. The algorithm incorporates a nonlinear proportional- proportional Integral controller in the inner loop, complemented by a proportional-integral (PI) controller in the outer loop. The experimental results show that the proposed cascade controller significantly enhances tracking accuracy while minimizing wear on the actuators in comparison to a PI controller lacking an inner loop. Jamroen et al. [127] proposed an alternative to LDR sensors using the ultraviolet spectrum. The results showed that the proposed system increases energy generation by 19.79% and 11% compared to the fixed flat-plate and LDR systems, respectively.

5.1.2.3. Hybrid. To enhance the overall tracking performance, a hybrid system is used. In these systems, structures from both open-loop and closed-loop tracking systems are integrated. Such an approach associates the use of open-loop for initial approximation and closed-loop for refinement. During the initial approximation stage, the open loop guesses the target's position and trajectory based on existing information, such as its last position or velocity. Once the target comes inside the range, the closed-loop system manages the control. Instantaneous data measured by sensors, cameras, or other measurement devices is recorded and compared with the predicted position from an open loop, calculating an error or residual. This error is then used to adjust the tracking parameters by control algorithms to minimize the inequality between the expected and measured positions.

Al-Othman et al. [128] developed a sun-tracking system designed to optimize the orientation of solar panels for the generation of electrical power from solar energy. To achieve this objective, a hybrid controller is implemented that combines an active search algorithm-based dynamic feedback controller with an open-loop solar monitoring system. During the testing phase, the solar panel's generated power increased by approximately 35%. Zhang et al. [129] devised a tip-tilt dual-axis hybrid tracking system. By employing photoelectric sensors to rectify installation deviation, the tracking system successfully aligns the solar altitude angle and azimuth angle with the solar trajectory. The proposed solar trajectory tracking algorithm has good accuracy in calculating hour angle, declination angle, altitude angle, and azimuth angle, with maximum relative calculation errors of 0.12°, 0.51°, 0.47°, and 0.91° respectively.

Safan et al. [130] introduced the design, modelling, and simulation of a two-axis Sun Tracking System equipped with a hybrid-controlled tracking system. The system exhibited a performance ratio ranging from 95 to 97%, resulting in a 23.3% power increase and a significantly faster dynamic response of 57.5% in tracking. Ferdaus et al. [131] presented the design and construction of a dual-axis hybrid solar tracking system. Its effectiveness was evaluated and compared with a dual-axis solar tracking system operating in both static and continuous modes. According to the results, the proposed system achieved a power gain of 25.62% more than the stationary panel and consumed approximately 44.44% less power than the dual-axis solar tracking system.

5.1.3. Semi-passive

For more optimization of the solar energy collection, Semi-passive tracking, or in other words modified passive solar tracking is used. This tracking system links passive and active tracking techniques. Also, it can maintain the sun's rays perpendicular to the absorbers by smoothly following the sun's movement. This is achieved with minimal mechanical intervention and reduced sun-tracking movement. Leon et al. [132] and [133] utilized a semi-passive tracking system which consists of a micro-heliostat array, a Fresnel lens, and a receiver. The microarray monitors the sun's position and forwards the sun's rays towards the horizontal lens. The kinematics of the semi-passive solar tracking concentrators are investigated to increase the effective area of the Fresnel lens by minimize shading. The authors concluded that concentrating solar radiation in a limited area of the receiver increases energy collection and storage efficiency, consequently enhancing the system's overall efficiency.

5.2. Degree of freedom (DOF)

The degree of freedom (DOF) is a significant parameter that is used to describe the tracking systems based on the number of directions that the systems rotated. Based on this parameter, there are two types of tracking systems, single-axis and dual-axis.

5.2.1. Single-axis (1-DOF)

Single-axis tracking systems have only one degree of freedom that allows the system to rotate around one axis of rotation. This axis either be horizontal to track tilt angle (elevation tracking) or vertical to track the azimuth angle (azimuth tracking). In azimuth tracking, the azimuth angle changes from east to west to track the sun's daily movement while

Table 5
Comparative study of solar tracking systems

Reference	Study type	Work target	Tracking type	Main findings
Clifford and Eastwood (2003) [117]	Theoretical and Experimental	Design a passive solar tracking system.	- Passive - Dual-axis	The efficiency increased up to 23%.
Natarajan and Srinivas (2016) [118]	Theoretical and Experimental	find the tracking wheel radius for a continuous, power-free tracker in a linear solar concentrator.	- Passive - Single axis	The tracking improved with an average value of 1.5%
Alemayehu and Admasu (2023) [119]	Theoretical and Experimental	propose a passive solar tracking system using a bimetallic strip deflector.	Passive	It also shows 24.86% higher energy-collecting efficiency and 96.4% accuracy compared to an ideal tracker.
Riyadi et al. (2023) [120]	Experimental	performance of the combined PV and TE generators with different solar tracking systems.	- Active - Fixed - Single axis - Dual-axis	- The maximum power increased by 84% originating from PV power. - The efficiency improved by 3% compared to the stand-alone system.
Pirayawaraporn et al. (2023) [121]	Theoretical and Experimental	Develop an innovative sensor-less tracker to address the limitations of existing sensor-less solar tracking systems.	- Open - Dual-axis	The energy generation was enhanced by 20.1% compared to the fixed plate system.
Mi et al. (2016) [122]	Theoretical	Develop an open-loop solar tracking strategy for HCPV/T systems by using variable tracking frequency.	- Open	- Saving energy - Improved stability
Fathabadi (2018) [123]	Experimental	Introduce a novel sensor-less dual-axis solar tracking system that achieves high accuracy.	-Open	The system boosts energy efficiency by 28.8–43.6%.
Jamroen et al. (2020) [124]	Theoretical and Experimental	Suggest the design of a low-cost automatic dual-axis solar tracker system.	- Closed - Dual-axis	Increasing electrical energy efficiency by 44.89%.
Garrido and Díaz (2016) [126]	Theoretical and Experimental	Propose a cascade control algorithm to achieve a sun tracking error for applying it in HCPV.	- Closed	-The Maximum Tracking Error is 0.134° for the PI and 0.014° for the proposed system.
Jamroen et al. (2021) [127]	Experimental	Proposed an alternative to LDR sensors.	- Closed - Dual-axis	The system increases energy generation by 19.79% and 11% in comparison to the fixed flat-plate and LDR system.
Safan et al. (2018) [130]	Theoretical and Experimental	Proposed a hybrid solar tracking system that can generate high-output power.	- Hybrid	The performance ratio of the system ranged from 95% to 97%, producing 23.3% power and dynamic response that was 57.5% faster in tracking.
Ferdaus et al. (2014) [131]	Experimental	Explain the process of designing a solar tracker that is energy efficient, ranging from a conventional mechanical single-axis to a hybrid dual-axis system.	- Hybrid	The system has a power gain of 25.62% more than the stationary panel and power consumption of about 44.44% less than the dual-axis solar tracking system.
Zhu et al. (2020) [134]	Experimental	Introduce a design for a single-axis solar tracker to optimize energy collection	-Single axis	Achieves an annual radiation incident of 96.4% compared to a dual-axis tracking system.
Batayneh et al. (2019) [135]	Theoretical and Experimental	propose a distinct single-axis solar tracker that follows the sun's path by actuating in the azimuthal panel only three times per day.	- Single axis	Saving in energy consumption, as it was about 91–94%.
Munanga et al. (2020) [136]	Theoretical and Experimental	developed an intelligent single-axis solar tracking system.	- Single axis - Passive	The increase in efficiency was 25% compared to a fixed PV panel.
Azam et al. (2024) [137]	Experimental	Improved the performance of PV system by introducing semi continuous solar tracking algorithm.	Single axis	The semi continuous tracking reduces energy consumption by 34% compared to continuous tracking.
Huang et al. (2022) [138]	Theoretical	Used machine learning to enhance the efficiency for wireless power transmission	Single axis	The system successfully achieved as the deference between expected and actual efficiency was 1% − 3%.
Saymbetov et al. (2021) [140]	Experimental	propose a dual-axis schedule tracker with an adaptive algorithm for the case of strong sun rays scattering by clouds and compare that system with a normal dual-axis schedule tracker.	- Dual-axis	The results showed an enhancement of the performance by 41% compared to the normal dual-axis schedule tracker and generated 66–69% power more than the normal dual-axis.
Alomar et al. (2021) [141]	Theoretical and Experimental	Compare the performance of different solar tracking systems (fixed, single axis, dual axis).	-Single axis - Dual-axis	The produced energy increased by 16.5% and 25.5% for single-axis and dual-axis tracking systems respectively.
Alamher and Alahmer (2022) [142]	Experimental	Developed dual axis tracking system to improve the output energy.	Dual axis	The system achieved a maximum efficiency of 32.2%.
Mamodiya and Tiwari (2023) [143]	Experimental	Improved efficiency by using different control method	- Dual axis - Closed loop	The efficiency of solar PV tracking system is increased by 12%.

HC = high-concentration; LDR = light-dependent resistor; PV = photovoltaic.

the tilt angle is still constant throughout the year. In contrast, the elevation tracking allows the panel to track the sun's changing tilt angle throughout the day but it does not account for the sun's movement from east to west. The main advantages of this system are its simplicity, cost-effectiveness and low maintenance system compared to Dual-axis solar tracking systems. Zhu et al. [134] presented a single-axis solar tracking system that optimizes energy collection by enhancing the tracking performance based on sun-earth interactions and a predicted radiational model. The results showed that this proposed system has overcome the previous single-axis tracking configurations and accomplishes an annual radiation incident of 96.4% compared to a dual-axis tracking system.

Batayneh et al. [135] proposed a distinctive single-axis solar tracking system that tracks the sun's path by adjusting the azimuthal panel only three times per day. The results were used to compare this unique system against a normal continuous single-axis tracker, revealing significant energy savings of about 91–94% with the panel moving in only three angle adjustments. Munanga et al. [136] developed an intelligent single-axis solar tracking system using a binary dominance matrix to maintain the sun's rays perpendicular to the panel. Experimental results showed a 25% increase in efficiency compared to a fixed PV panel, which was a noteworthy improvement considering the typical range of existing systems being 15–20%. Azam et al. [137] improved the performance of PV system by introducing semi continuous solar tracking algorithm. The average improvement in short circle current was 10.01% for continuous tracking and 9.37% for semi continuous tracking PV system compared to fixed PV system. Huang et al. [138] designed a wireless power transmission that maximize efficiency by machine learning. The system successfully achieved as the difference between expected and actual efficiency was 1–3%.

5.2.2. Dual-axis (2-DOF)

Dual-axis solar tracking systems have two degrees of freedom that allow panels to rotate in both directions of azimuth and elevation. These systems are more complex and expensive compared to single-axis solar tracking systems, but they propose more accurate sun tracking kinematics because they adjust the orientation of the panel to track the sun's rays' path from sunrise to sunset. Additionally, they interpret the seasonal shifts in the sun's elevation angle throughout the year. Dual-axis solar tracking systems are used in applications where maximum electrical production and high performance are essential, such as HCPV systems.

Muthukumar et al. [139] introduced a dual-axis solar tracker to maximize the energy generation from the panel. The panel's positioning is controlled by a microcontroller that responds to the sun's movement. With a rapid 0.2-second response time, the system efficiently adjusts to ensure continuous power generation. The direction of movement changes when there is 0.8 V difference between light dependent resistor outputs. Saymbetov et al. [140] proposed a dual-axis schedule tracker with an adaptive algorithm specifically designed to handle the strong scattering of sun rays caused by clouds. The proposed system showed an enhancement of the performance by 41% compared to the normal dual-axis tracker resulting in power generation of 66–69%. Alomar et al. [141] conducted an analysis of the energetic, economical, and environmental aspects of a PV solar system utilizing different tracking systems under climatic conditions in the north of Iraq. The study revealed an increase in energy production of 16.5% and 25.5% for single-axis and dual-axis tracking systems, respectively. Table 5 represents a review of some of the studies that investigate solar tracking systems. Al-Amayreh and Alahmer [142] developed dual axis tracking system to enhance energy output to produce hot water and light as the experimental work achieved a maximum efficiency of 32.2%. Mamodiya and Tiwari [143] used a different control method as sensors feedback controller that compared the sun light intensity with the threshold to improve efficiency. The tracking simulation shows increase in efficiency of solar PV tracking system by 12%.

6. Future challenges

Based on the outcomes of this manuscript, there are various areas that need more further investigation and these will be help in enhancing the PV systems performance. The following suggestions are offered for future research:

- Investigate novel passive cooling methods that employ enhanced materials or innovative designs to increase heat dissipation and therefore improve PV performance. Also in the active cooling systems, using liquid base cooling or PCM with good heat transfer properties to avoid the effects of high temperatures and increasing the PV life time. Additionally conducting qualified studies to estimate the usefulness and cost-efficiency of the different strategies of PV cooling under many environmental conditions.
- Explore innovative methods for spectral splitting to adjust the employment of different wavelengths of sunlight and improve overall performance. Also Inspect using advanced optical materials to improve spectral splitting performance. Integrating spectral splitting technique with present PV technologies, such as multi-junction solar cells.
- Examining new tracking algorithms to increase the accuracy and efficiency of solar tracking systems. Also discussion the integration of the sensors with the machine learning algorithms to control the system performance in the unpredictable conditions. Finally evaluate the economic viability and practicality of using tracking technology for the large scale PV farms.

By concentrating research efforts on these suggested fields, developments can be made for the PV systems and further enhance their performance.

7. Conclusion

A comprehensive review of the advancements in PV/T systems and their different variants and configurations was carried out. The study has shed particular light on four main modification methods related to cooling approaches, concentration strategies, spectral spelling filters, and solar tracking systems. The main remarkable points can be summarized as follows:

1. By reducing the PV panel temperature, PV/T technology has appeared as a feasible technique to address these challenges and enhance electrical productivity, thermal energy gained, and affordability.
2. Both active and passive cooling strategies have been investigated to alleviate the minus effects of temperature on the panels' efficiency. Both techniques employ gas, liquid, and PCM as coolant media. The amalgamation of PCMs with the active cooling system achieves better results in maintaining the PV panels' temperature. The employment of such cooling systems can yield remarkable benefits in terms of the system's performance or panel lifespan.
3. The use of concentrators such as flat concentrators, compound parabolic concentrators, and Fresnel lenses increases the radiation intensity as a result could replace a considerable number of PV panels. Consequently producing more energy and enhancing the system performance, especially by integration with PV/T systems. Furthermore, the cost-effectiveness of various levels of concentration, from low to high, appeared in the savings that may be realized through the installation of fewer PV panels.
4. The main concept of spectral splitting filtering techniques is to selectively divide the sun rays into specific ranges of the rays' wavelength, allowing for the efficient conversion of desired wavelengths into electricity while minimizing the negative impact of excessive heat.
5. There are two ways to achieve the spectral splitting technique, the reflective method and the absorptive method. The reflective method

is based on redirecting the incident rays by utilizing thin-film wave interference filters or selective absorbers, while the absorptive method prevents undesired wavelengths from reaching the PV panel by employing materials with selective absorption properties

6. Various tracking mechanisms could be used to optimize the alignment of incident rays with the PV panel. These tracking mechanisms guarantee that the PV panel receives the maximum quantity of sun rays throughout the day, leading to enhanced energy conversion. The integration of the tracking systems with the PV panels not only enhances the PV electrical efficiency but also contributes to increased power generation and overall system energy.

To sum up, this comprehensive study offers insightful information on updated advancements in PV systems, emphasizing the possibility of increasing energy production, system overall efficiency, and affordability. The results encourage continued efforts to produce zero-emission and clean electricity to cover energy demand worldwide and fight climate change. Consequently, the contribution of solar energy to global electricity generation could be increased by implementing the proposed modifications, including cooling methods, concentration techniques, spectral splitting filtering, and tracking systems. This will pave the way for a sustainable and environmentally friendly energy future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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